ACTIVE DIRECTORY LAB

Windows Server Domain Controller + Windows 10 Client

Overview

Configuration of an Active Directory environment, including the installation and promotion of a Windows Server as a domain controller, creation of the `Mydomain.local` domain, management of users, groups, and Organizational Units (OUs), as well as the configuration of Group Policies (GPOs). Integration of a Windows 10 client into the domain, network/DNS configuration, and validation of domain user authentication.

Lab Workflow

1. Configure the Windows Server VM and network.
2. Install Active Directory Domain Services.
3. Promote the server to a Domain Controller and create the Mydomain.local forest.
4. Create OUs and an Active Directory security group.
5. Configure Group Policy settings.
6. Configure the Windows 10 client network and computer name.
7. Join the client to Mydomain.local.
8. Verify PC1 in Active Directory and log in as User01.

Step 1 — Configure the Windows Server VM Network Adapter

What I did: I configured the Windows Server virtual machine network adapter in VirtualBox and attached it to the internal network used for the Active Directory lab.

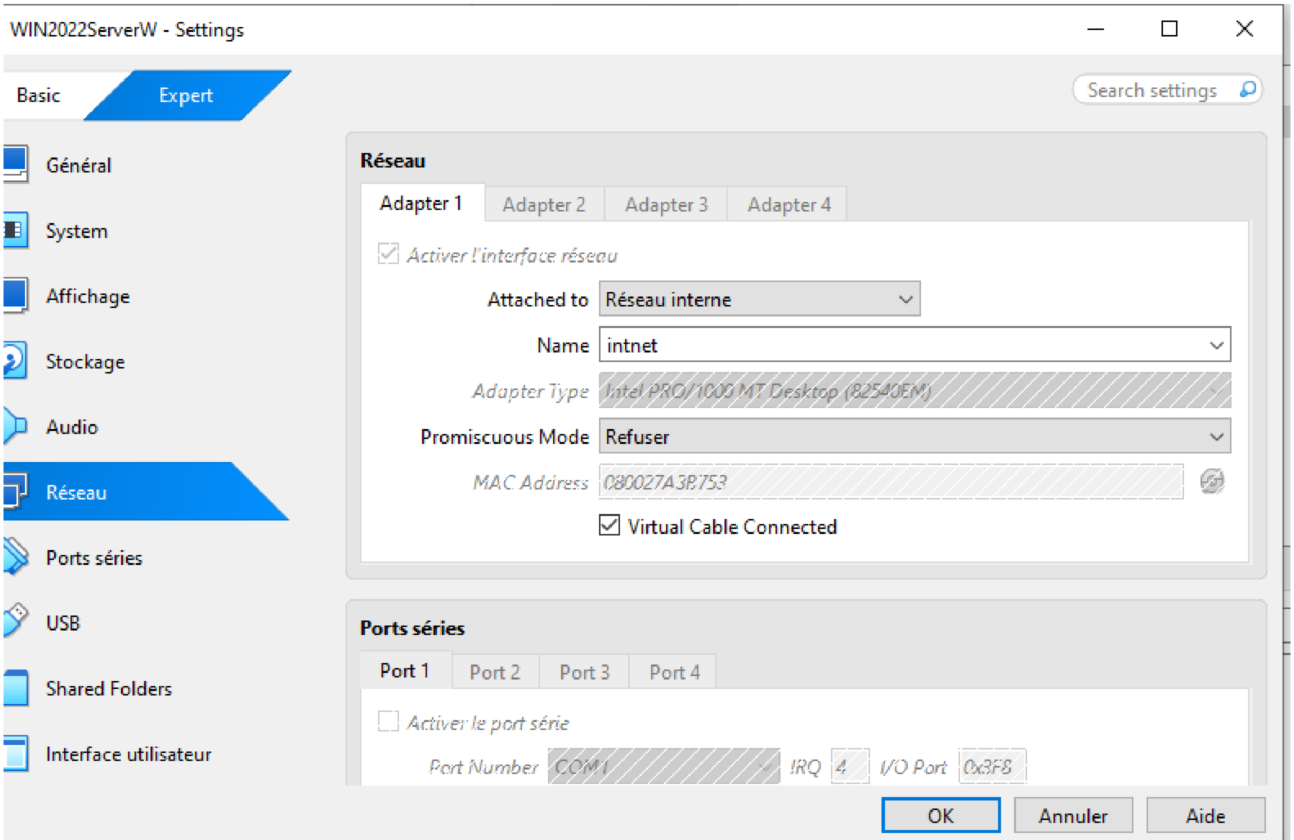

Figure 1: Configure the Windows Server VM Network Adapter

Step 2 — Configure a Static IPv4 Address and DNS

What I did: I opened the Ethernet IPv4 properties and configured the server with static network settings. The screenshot shows a static address configuration and DNS pointing to the server/local DNS service.

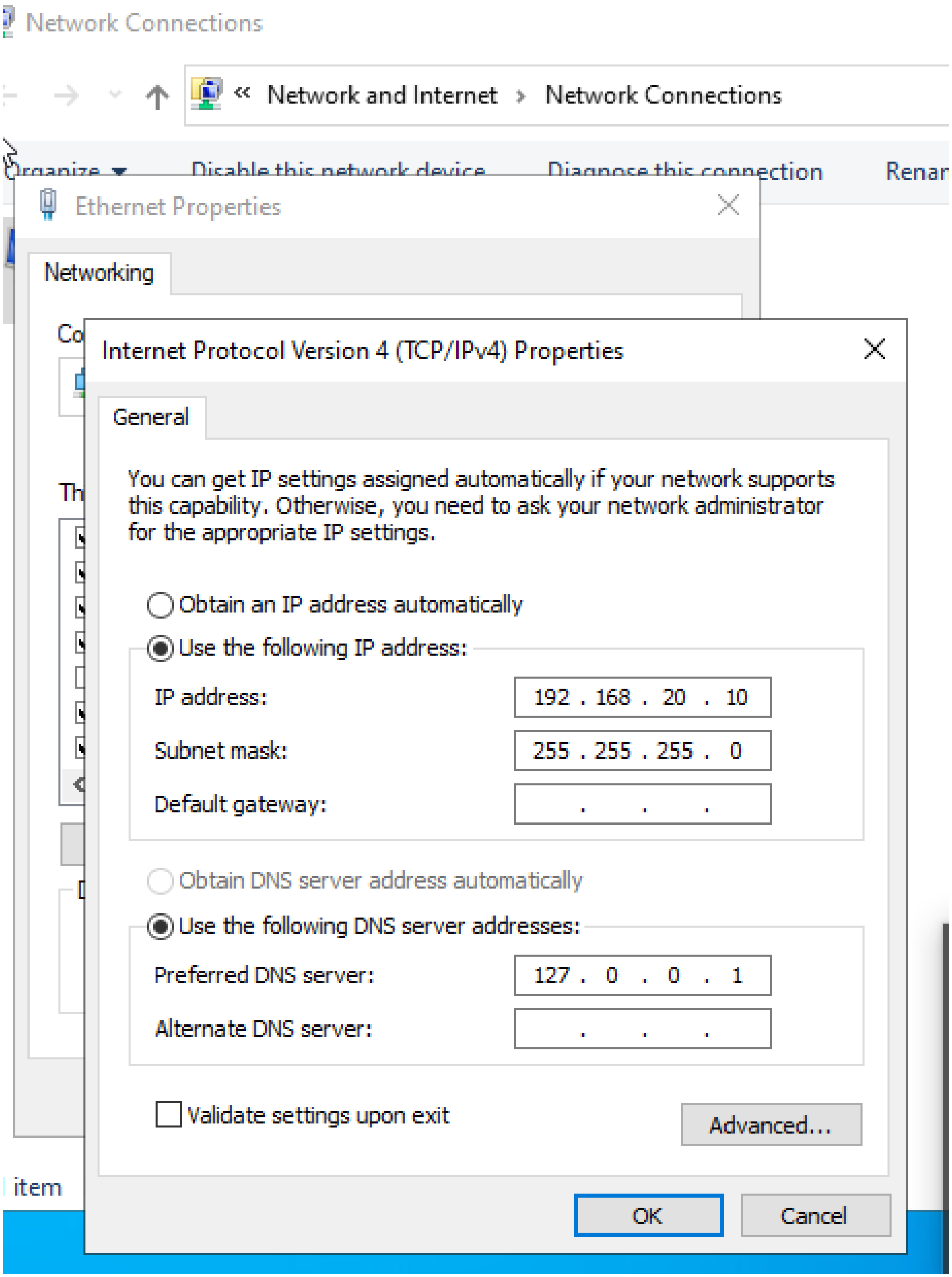

Figure 2: Configure a Static IPv4 Address and DNS

Step 3 — Rename the Windows Server

What I did: I started the Windows Server rename procedure so the server could have a meaningful computer name instead of the automatically generated name.

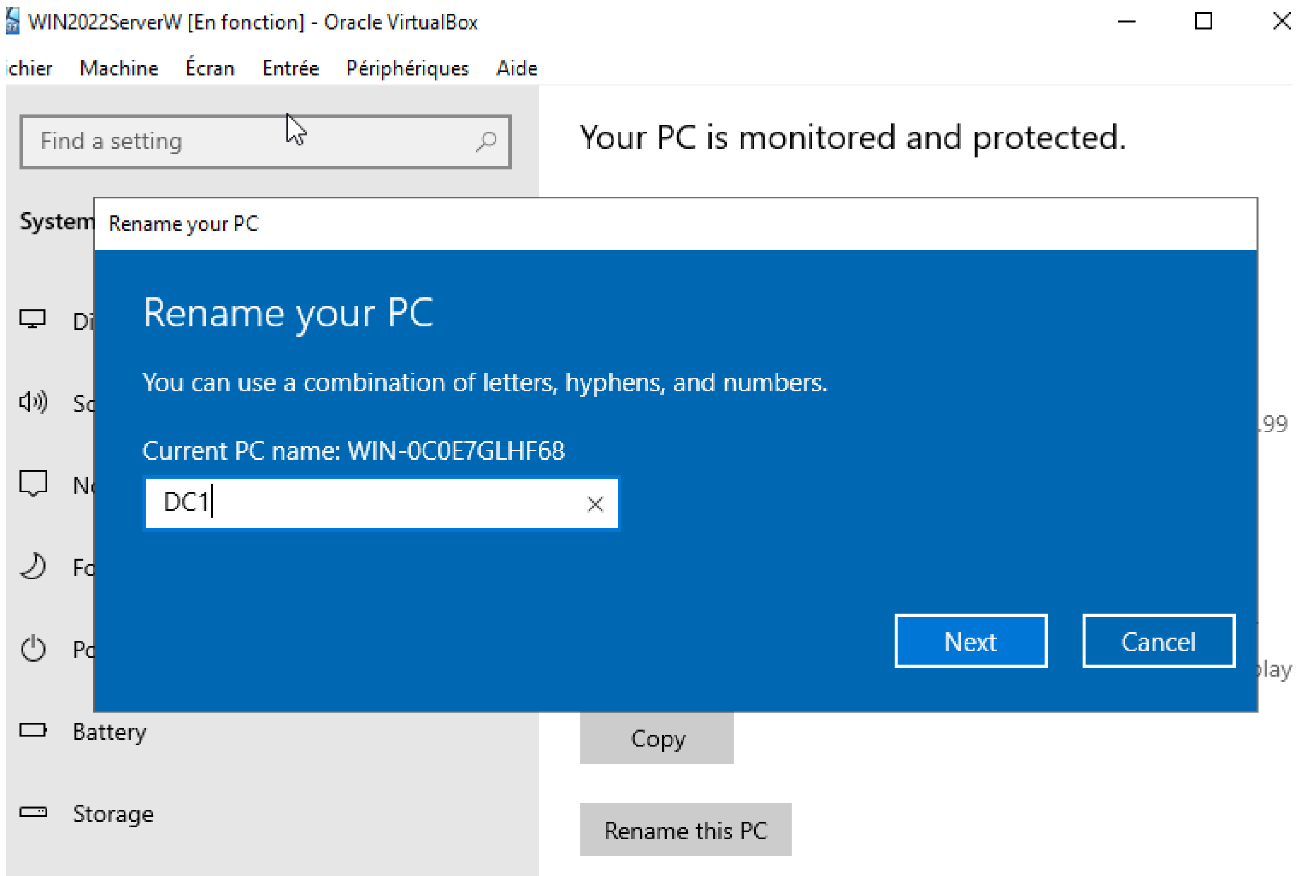

Figure 3: Rename the Windows Server

Step 4 — Open Server Manager and Configure the Local Server

What I did: I opened Server Manager and used the local-server configuration area as the starting point for installing the Active Directory role.

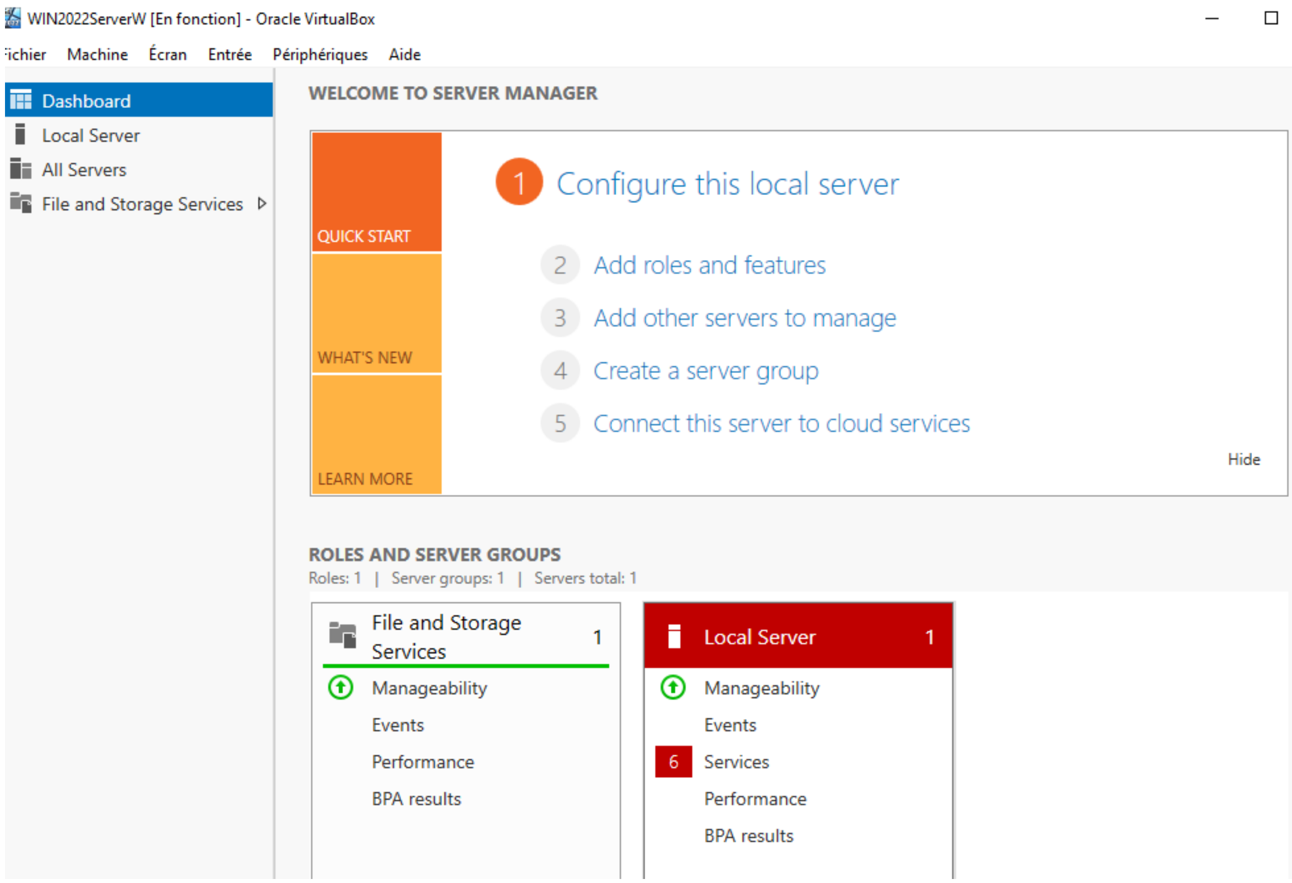

Figure 4: Open Server Manager and Configure the Local Server

Step 5 — Add Roles and Features — Select the Target Server

What I did: I launched the Add Roles and Features Wizard and selected the Windows Server from the server pool as the target for the installation.

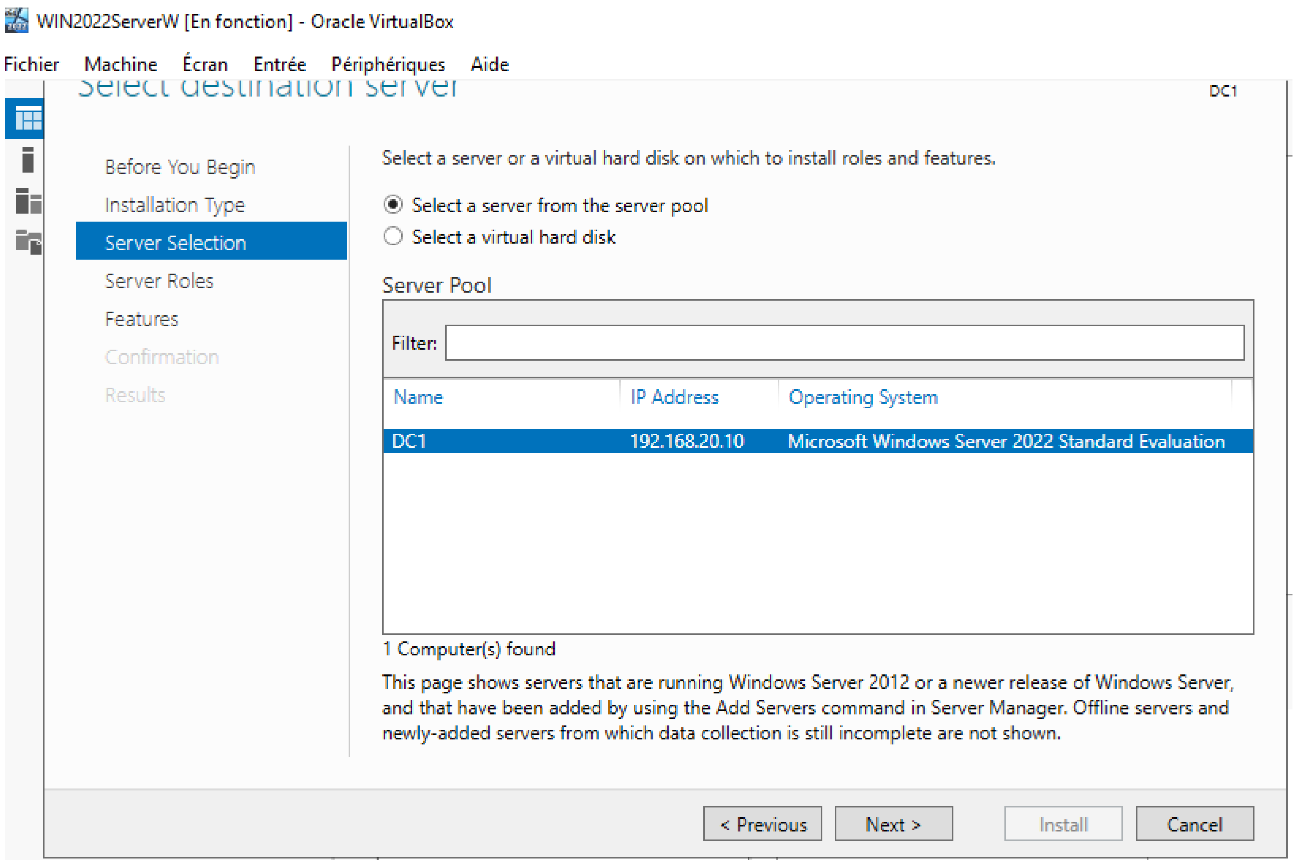

Figure 5: Add Roles and Features — Select the Target Server

Step 6 — Select the Active Directory Domain Services (AD DS) Role

What I did: I selected Active Directory Domain Services in the Server Roles page. This installs the role required to turn the server into an Active Directory Domain Controller.

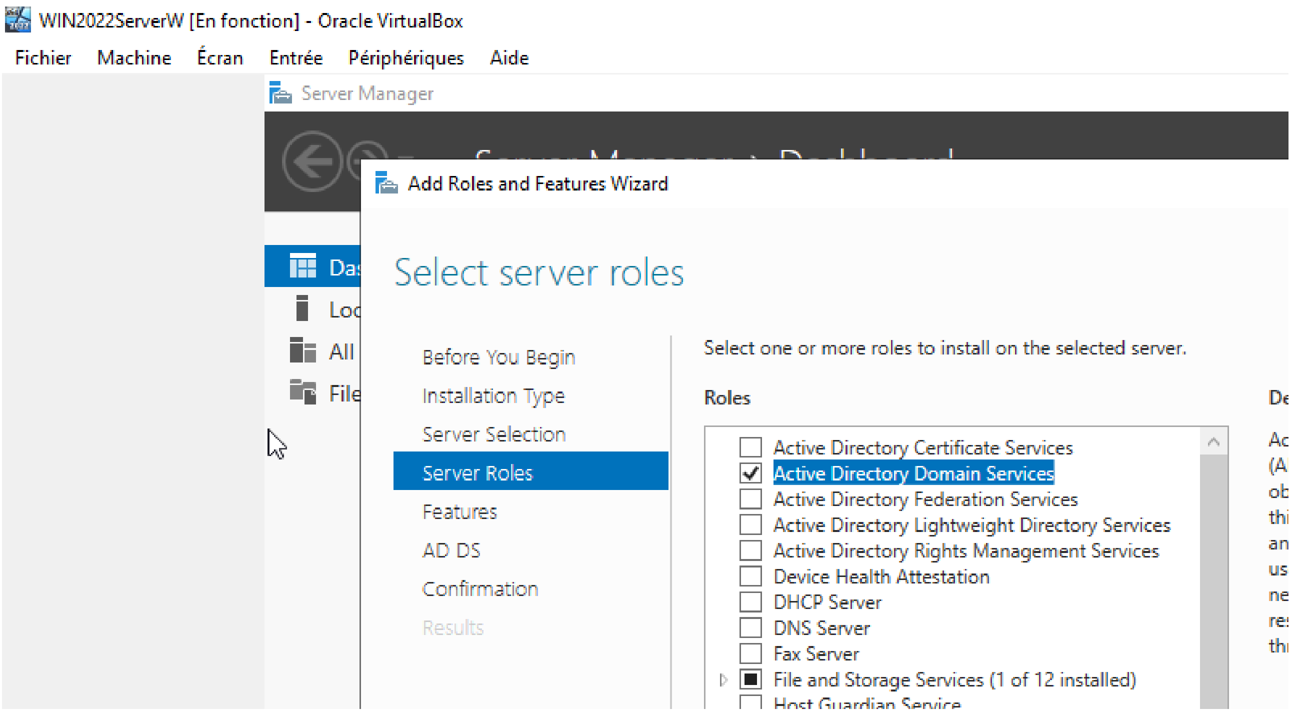

Figure 6: Select the Active Directory Domain Services (AD DS) Role

Step 7 — Install the AD DS Role and Required Features

What I did: I confirmed the role installation and allowed Windows Server to install Active Directory Domain Services and its required management tools.

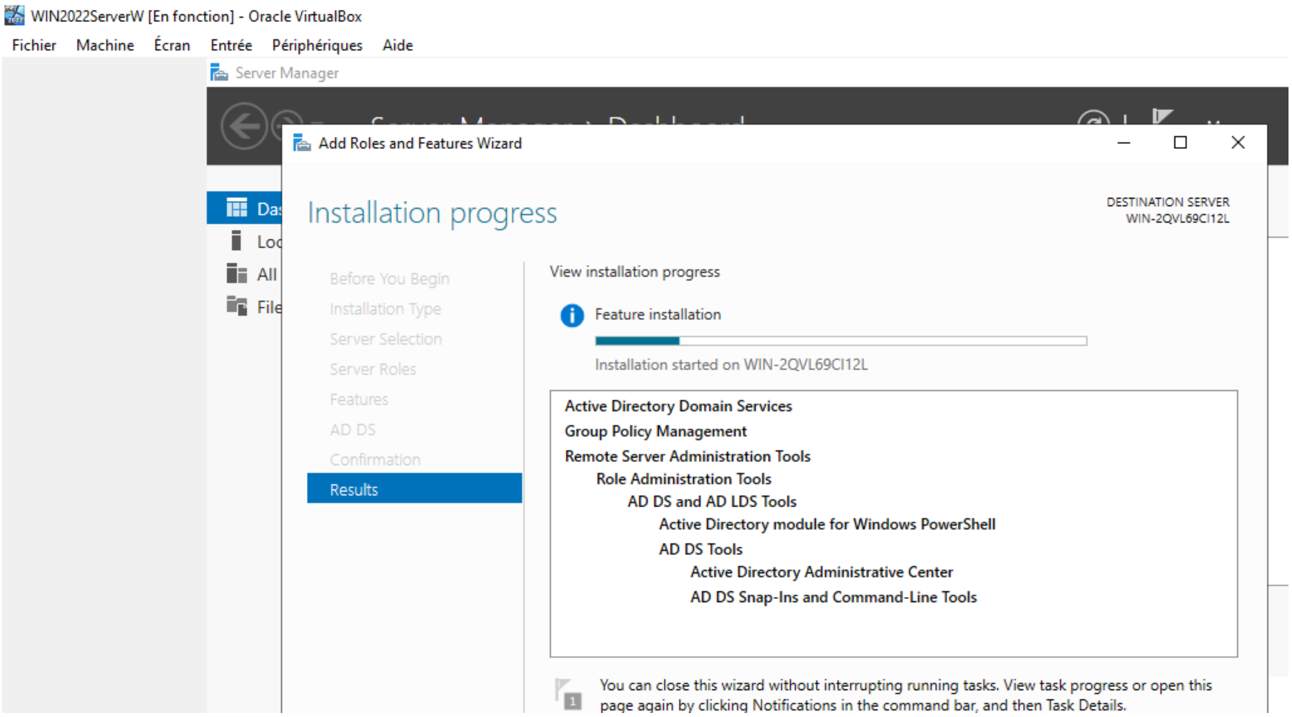

Figure 7: Install the AD DS Role and Required Features

Step 8 — Configure the New Active Directory Forest

What I did: I opened the Active Directory Domain Services Configuration Wizard and selected the option to add a new forest. The root domain shown in the screenshot is Mydomain.local.

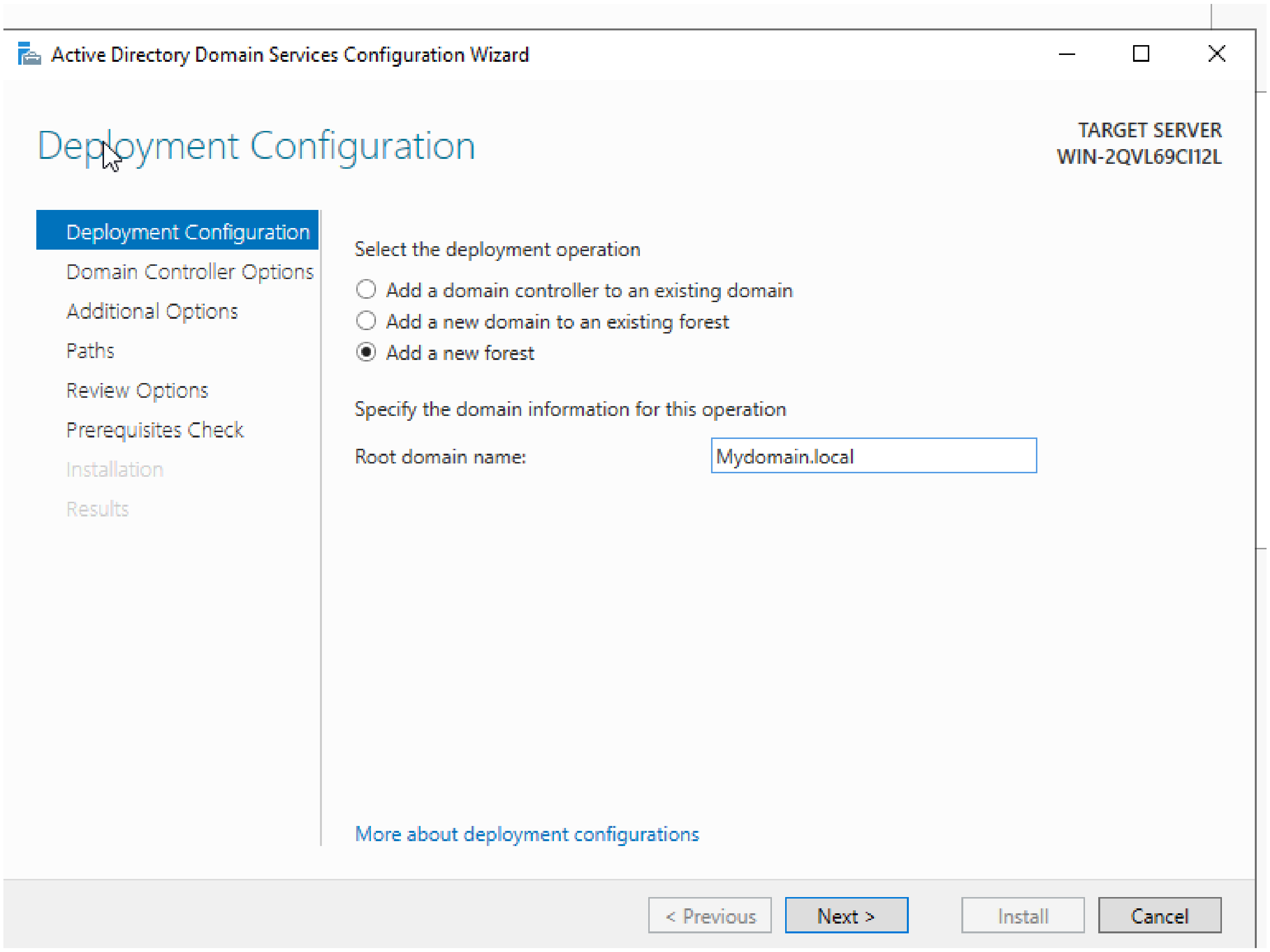

Figure 8: Configure the New Active Directory Forest

Step 9 — Configure Domain Controller Options

What I did: I configured the Directory Services Restore Mode password.

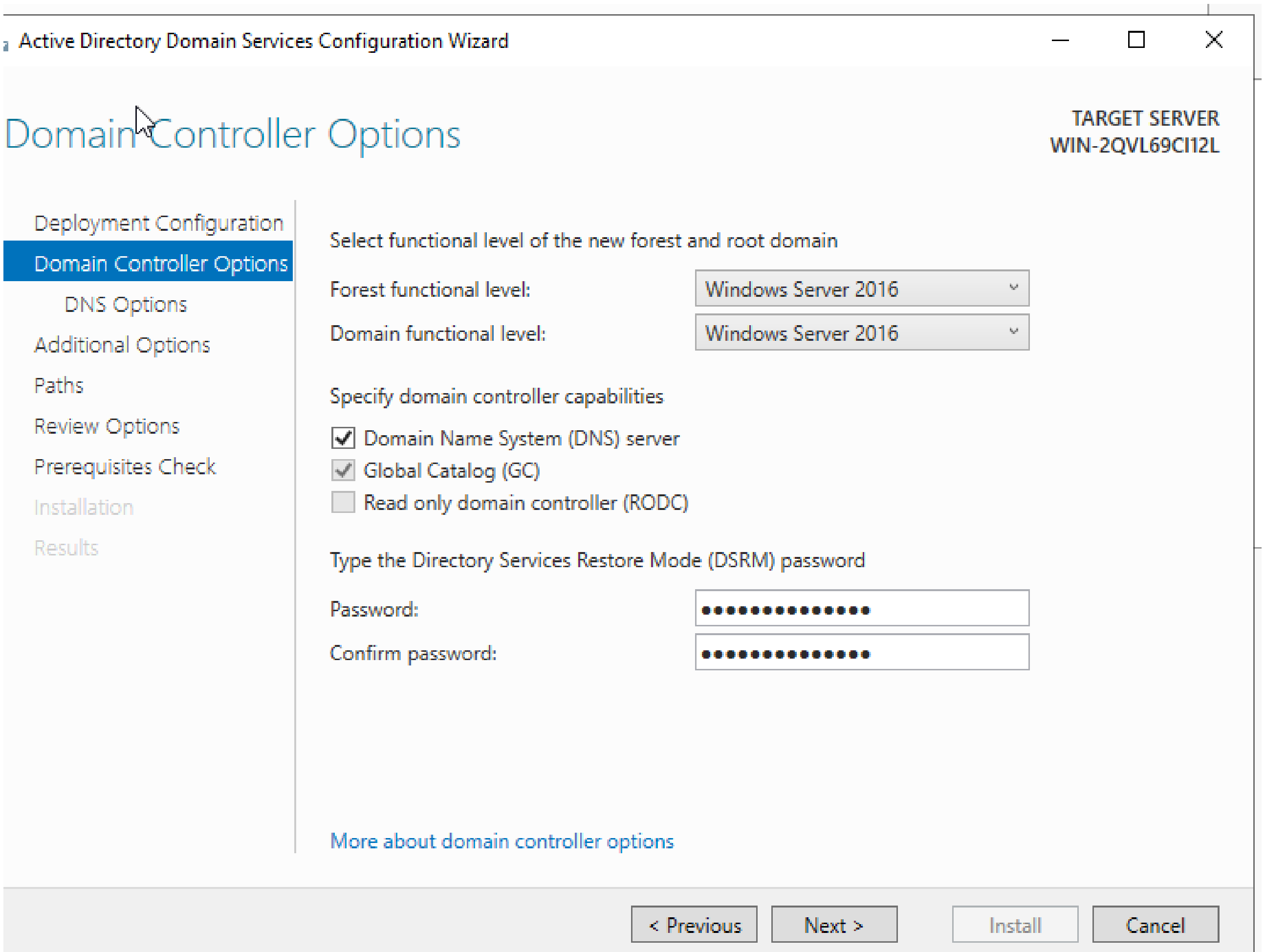

Figure 9: Configure Domain Controller Options

Step 10 — Complete the Domain Controller Installation

What I did: I started the AD DS configuration and waited for the Domain Controller installation to complete.

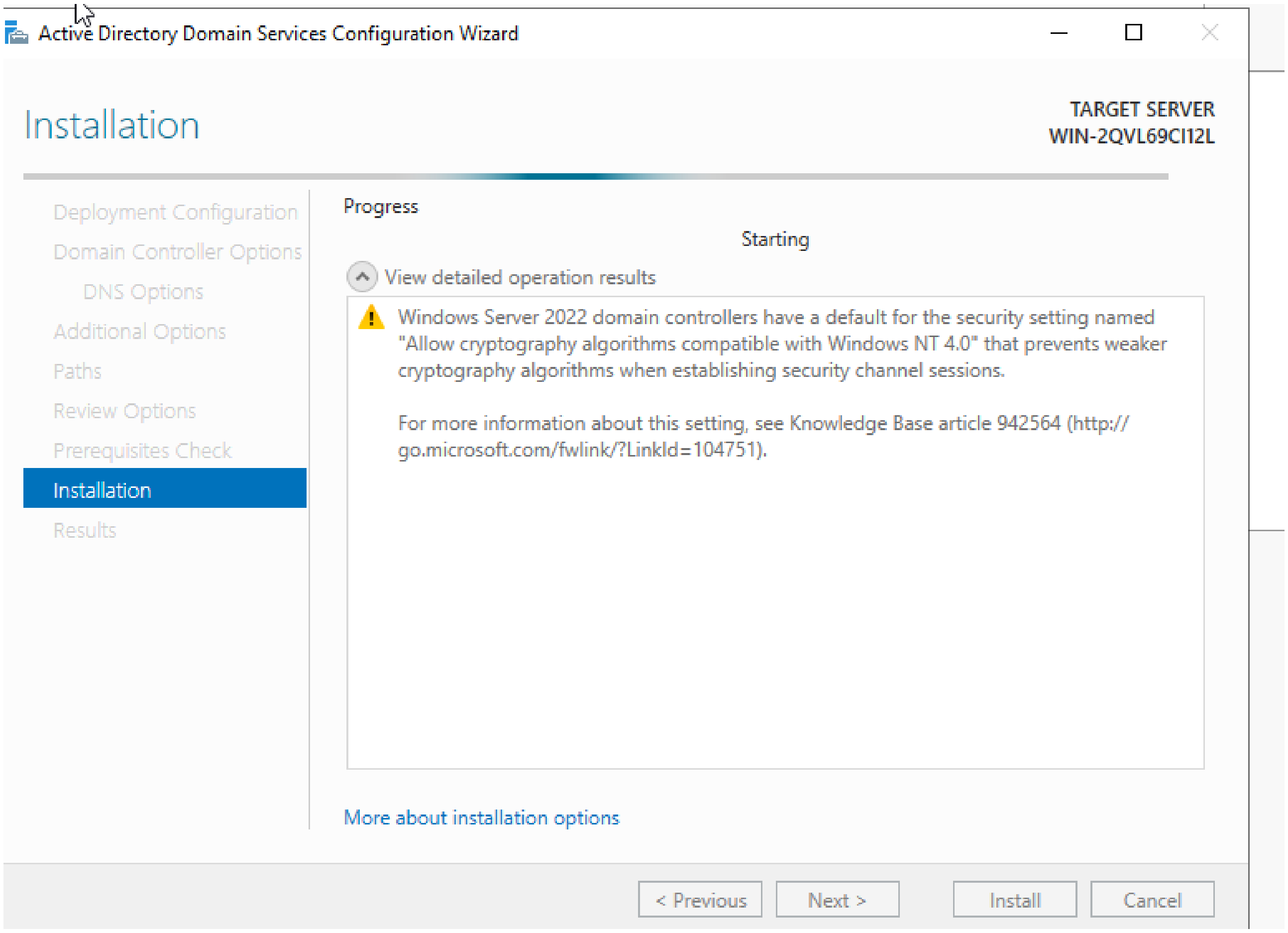

Figure 10: Complete the Domain Controller Installation

Step 11 — Log in Using the New Domain Administrator Account

What I did: After promotion, I logged in using the newly created domain and the domain Administrator account. The screenshot shows the MYDOMAIN domain context.

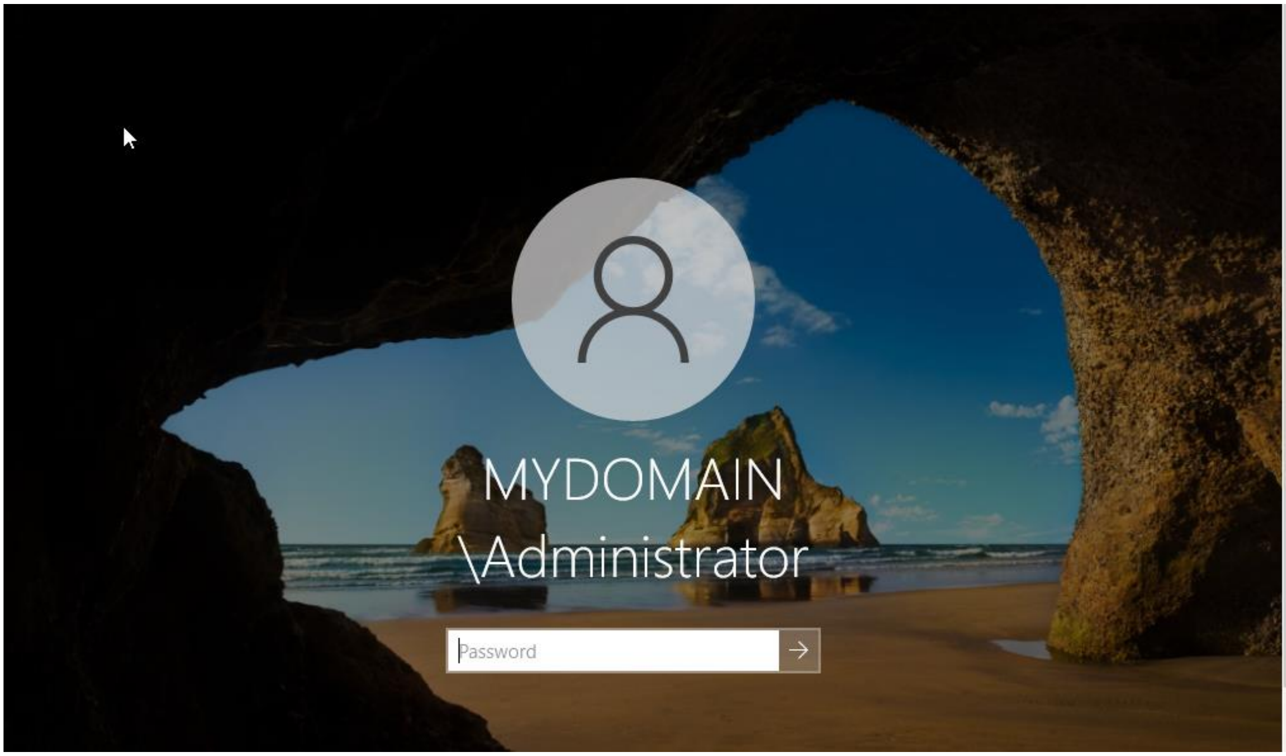

Figure 11: Log in Using the New Domain Administrator Account

Step 12 — Create a New Organizational Unit (OU)

What I did: In Active Directory Users and Computers, I opened the New Object menu and selected Organizational Unit to create a new OU under the domain.

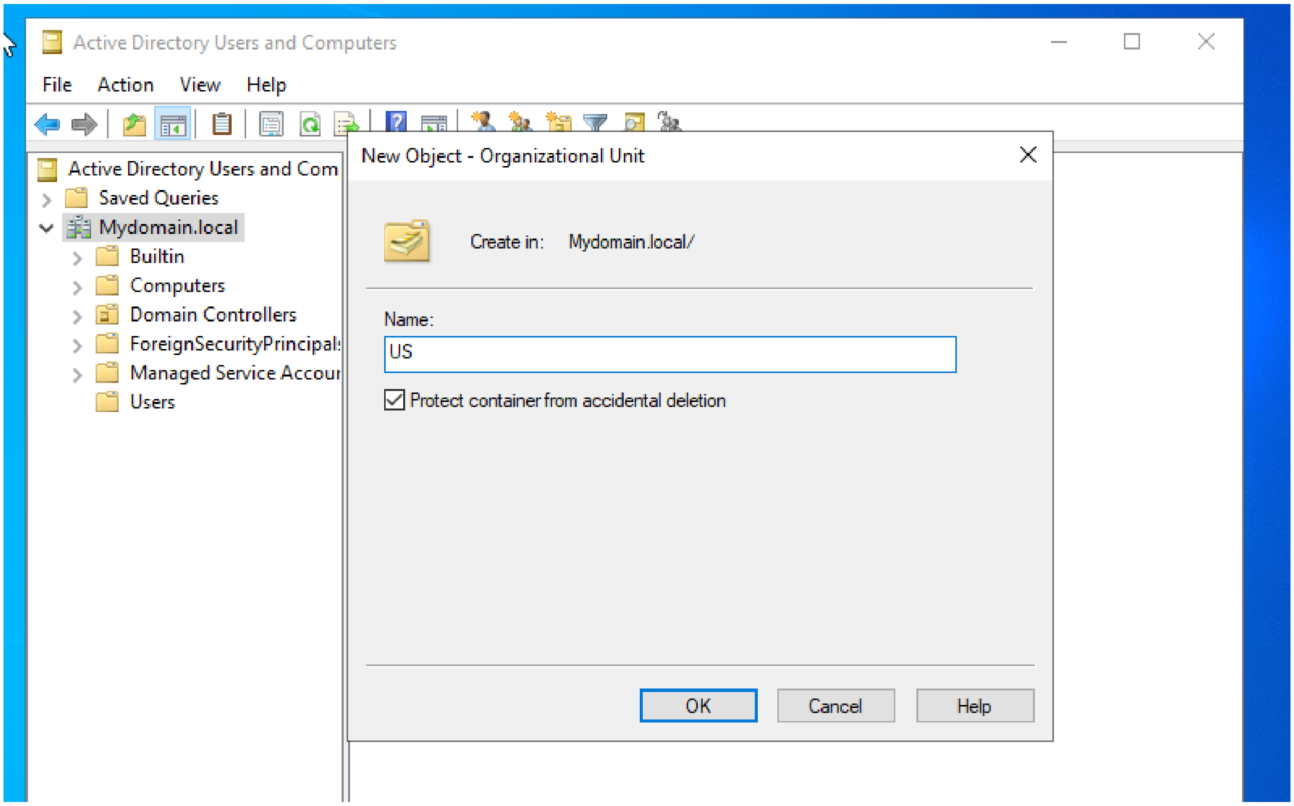

Figure 12: Create a New Organizational Unit (OU)

Step 13 — Verify the Organizational Unit Structure

What I did: I verified the Active Directory structure and the newly created organizational units. The screenshots show OUs such as Asia and Europe with separate computer, server, and user containers.

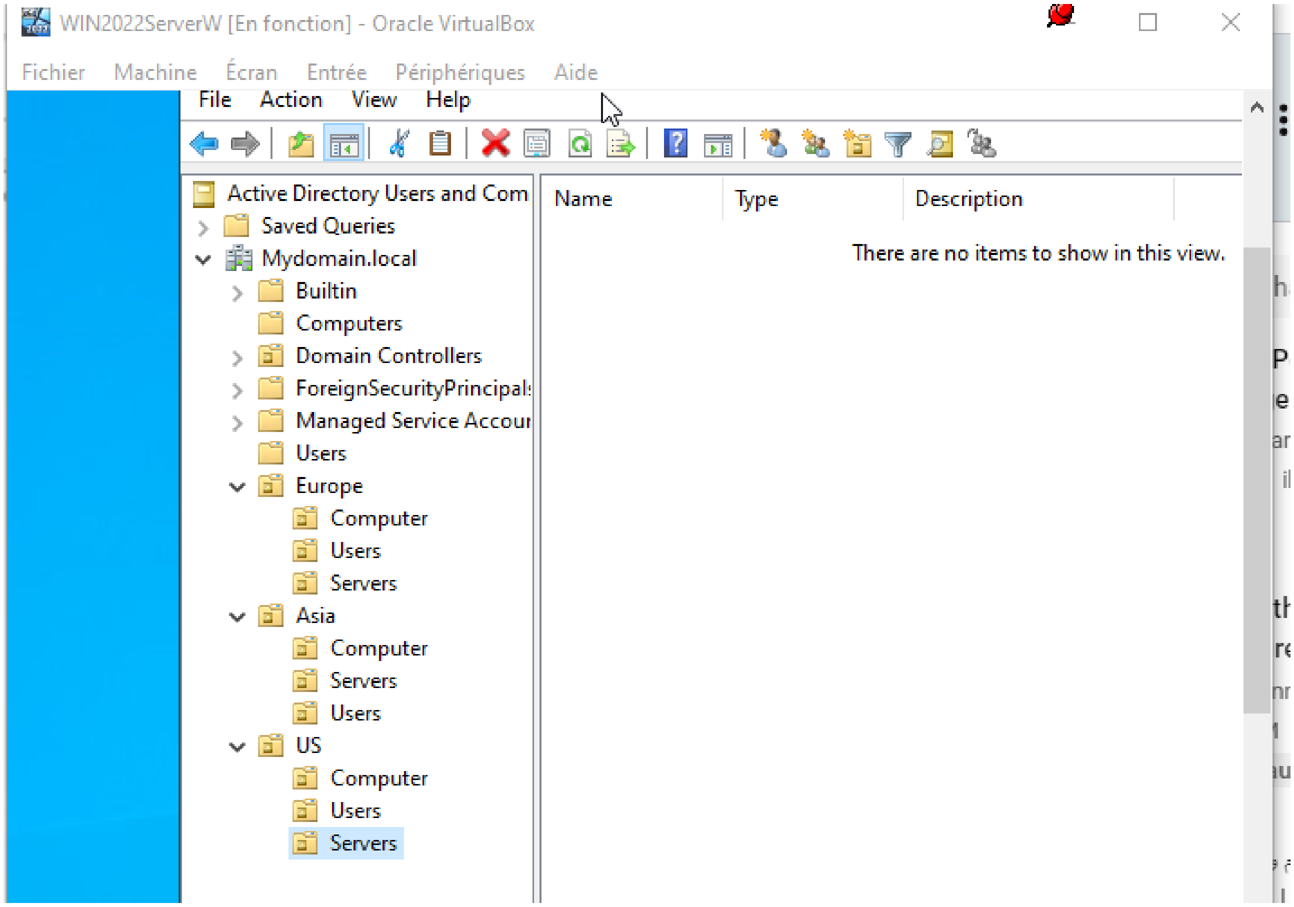

Figure 13: Verify the Organizational Unit Structure

Step 14 — Create a Security Group

What I did: I created a new group and user in Active Directory Users and Computers. The dialog shows the group name being entered and Security selected as the group type.

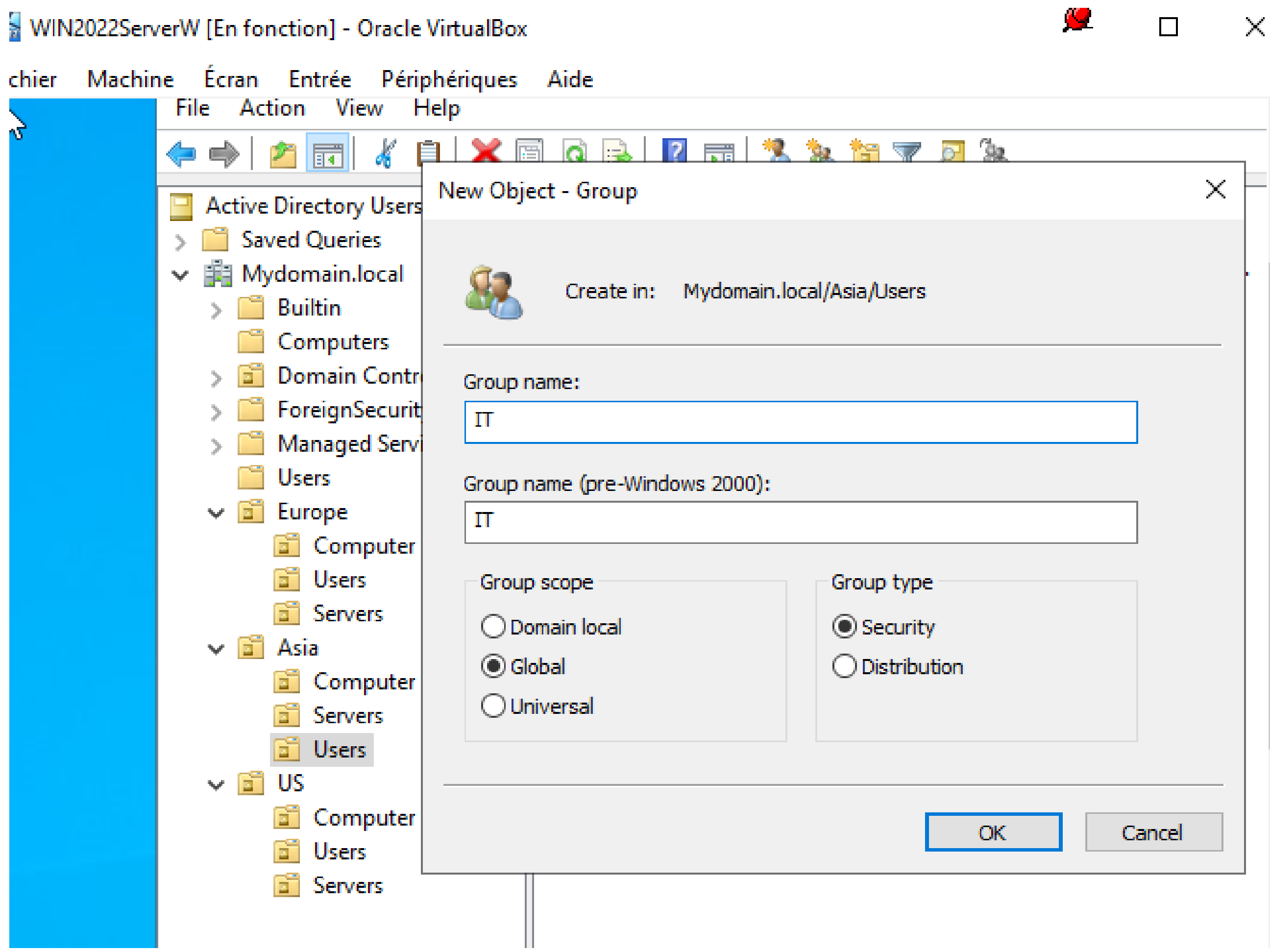

Figure 14: Create a Security Group

Step 15 — Verify the Active Directory Group

What I did: I returned to Active Directory Users and Computers and verified that the created group and user is present in the selected organizational unit.

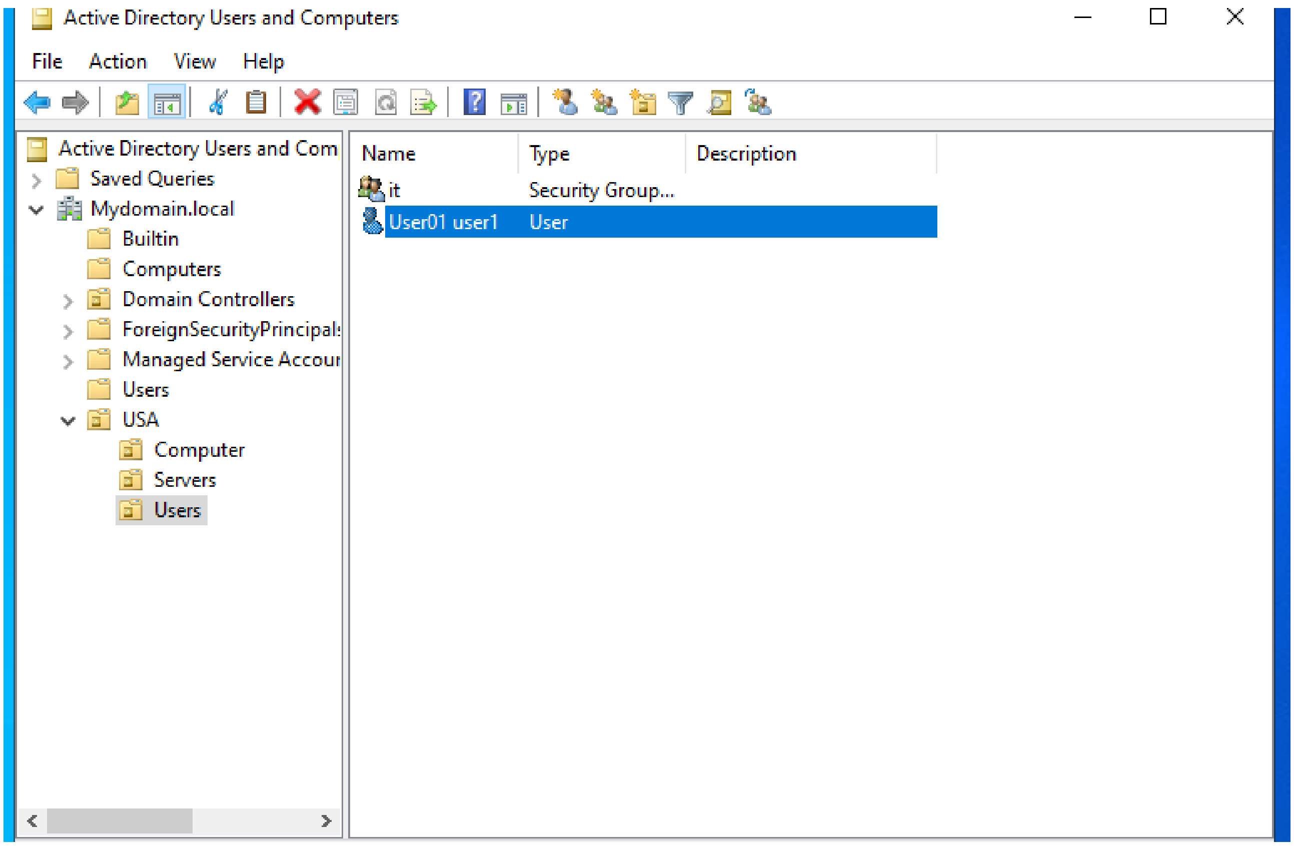

Figure 15: Verify the Active Directory Group

Step 16 — Create a Domain Group Policy Object for Password Policy

What I did: In Group Policy Management, I created/configured a Group Policy Object for the domain and worked on the Password Policy settings.

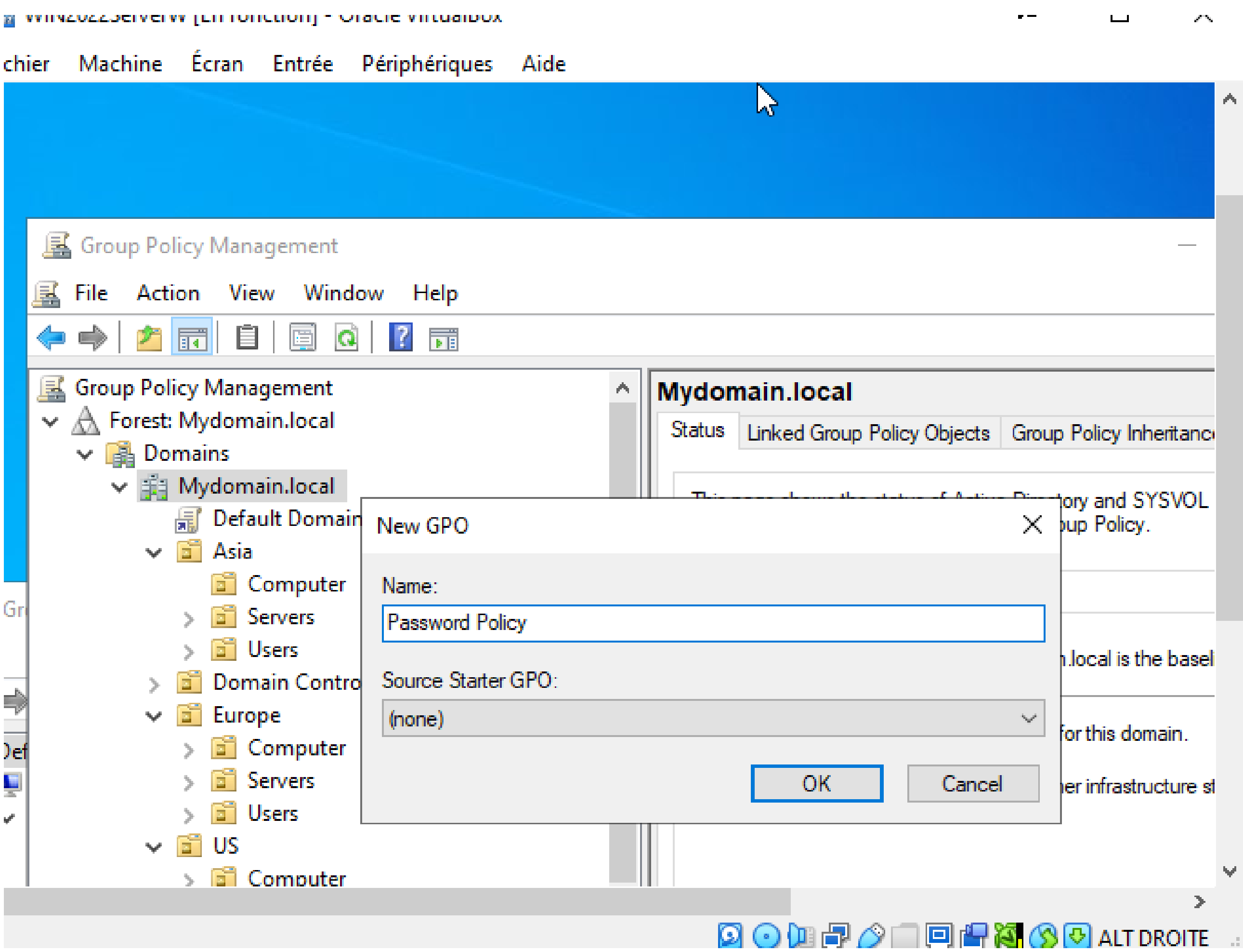

Figure 16: Create a Domain Group Policy Object for Password Policy

Step 17 — Configure the Minimum Password Length

What I did: I edited the password policy and enabled the Minimum password length setting, configuring the policy to require at least 12 characters.

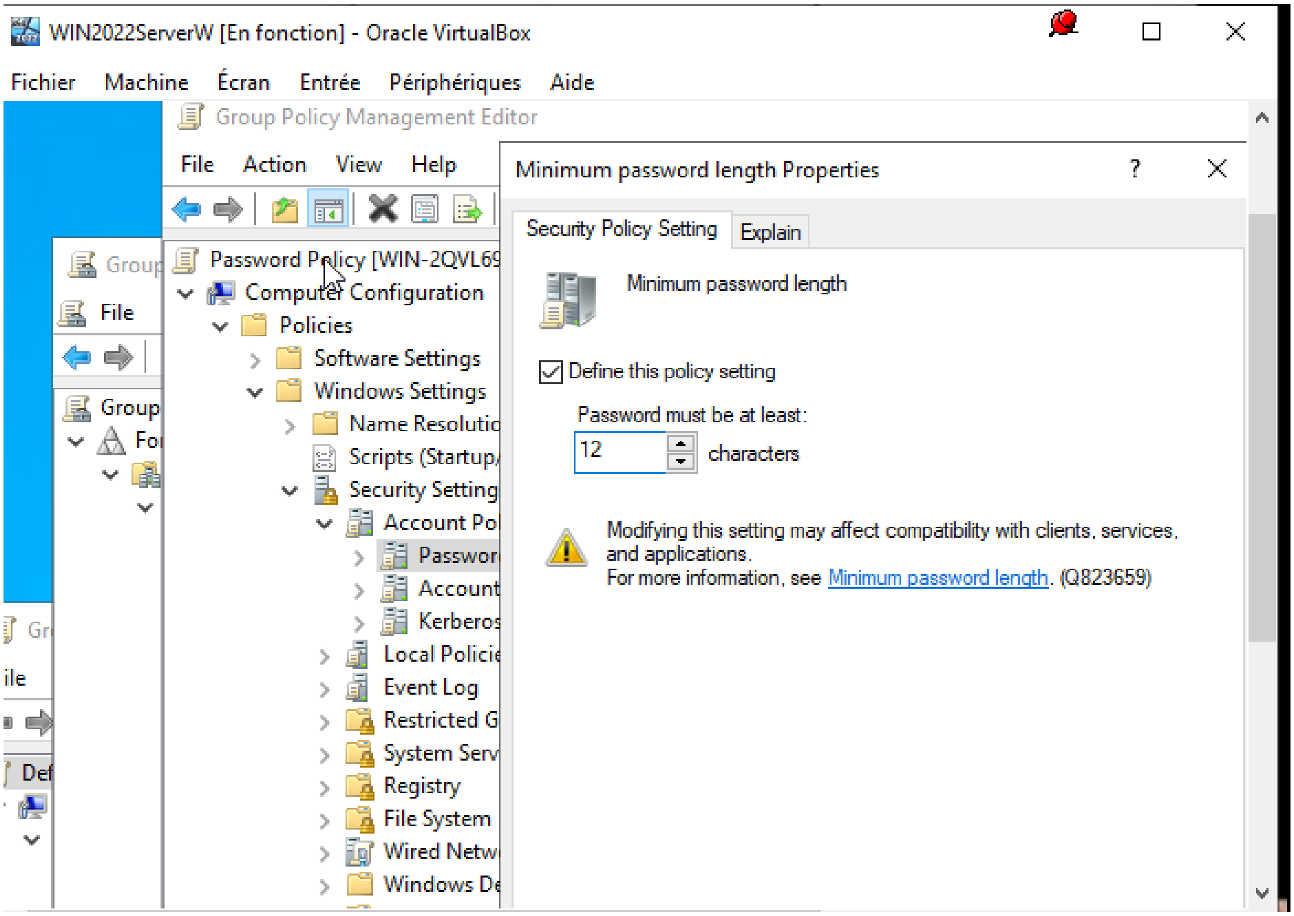

Figure 17: Configure the Minimum Password Length

Step 18 — Configure the Maximum Password Age

What I did: I configured the password policy's Maximum password age setting in the Group Policy Management Editor.

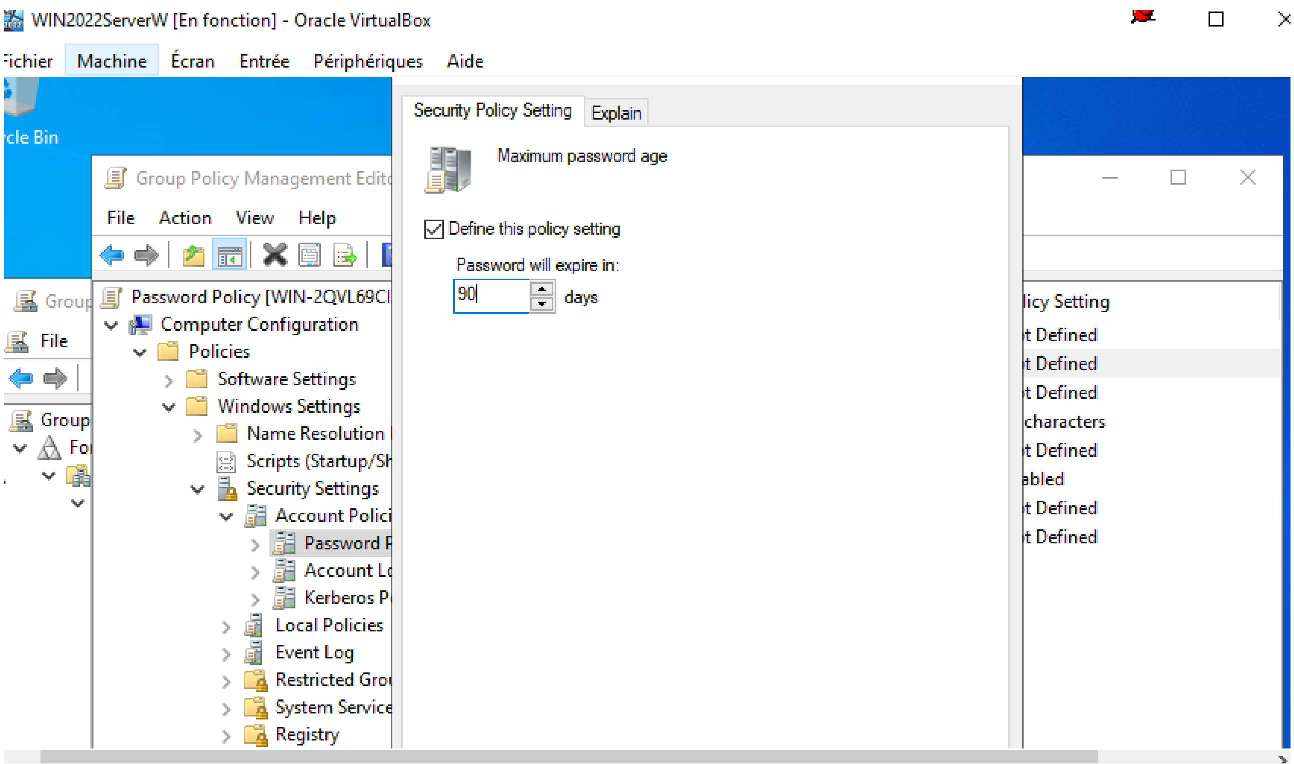

Figure 18: Configure the Maximum Password Age

Step 19 — Create a Drive Mapping Group Policy

What I did: I created a Group Policy Object for drive mapping under the domain and opened its configuration.

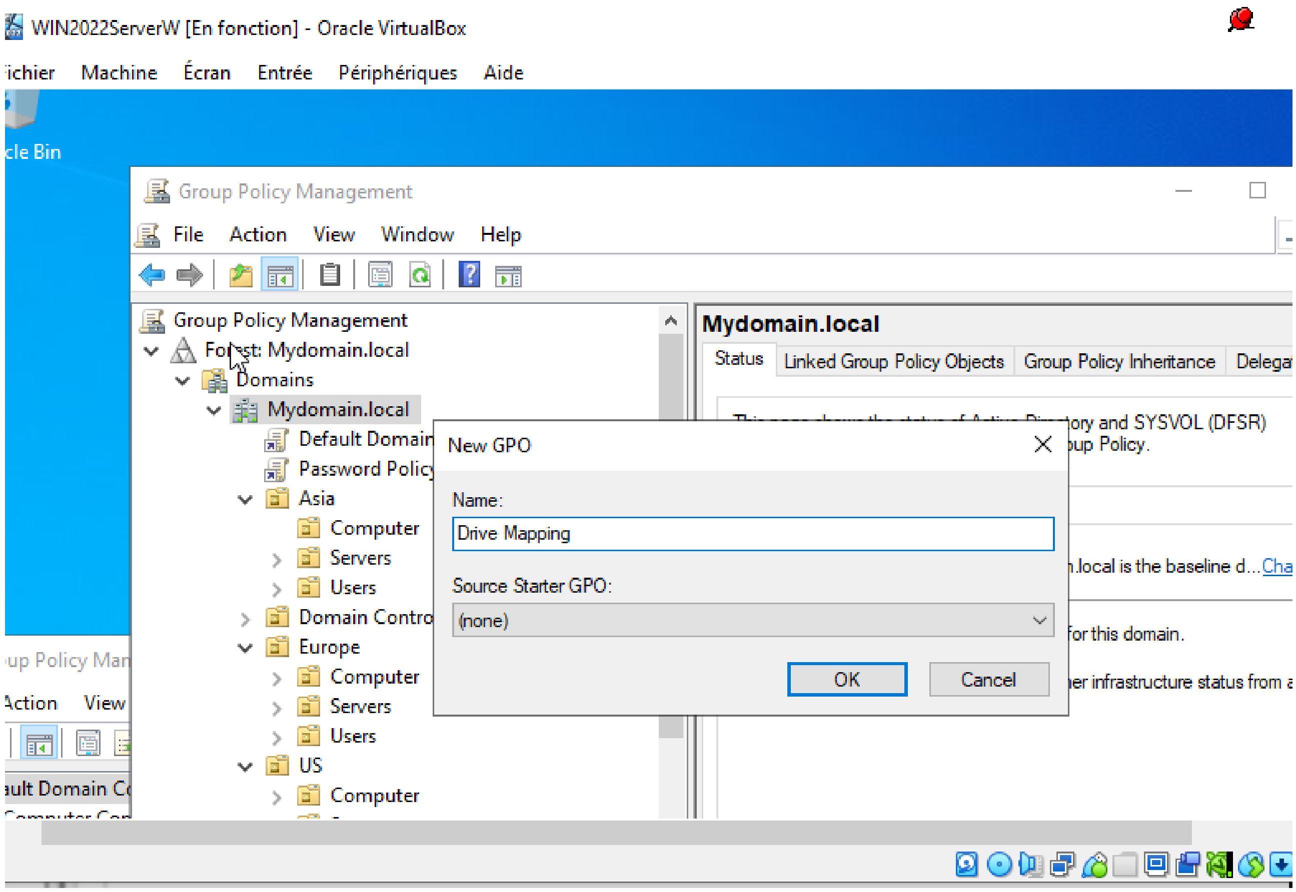

Figure 19: Create a Drive Mapping Group Policy

Step 20 — Configure the Drive Mapping Policy

What I did: I configured the drive-mapping policy so a network drive can be mapped for users. The screenshot shows the drive letter and mapping action settings.

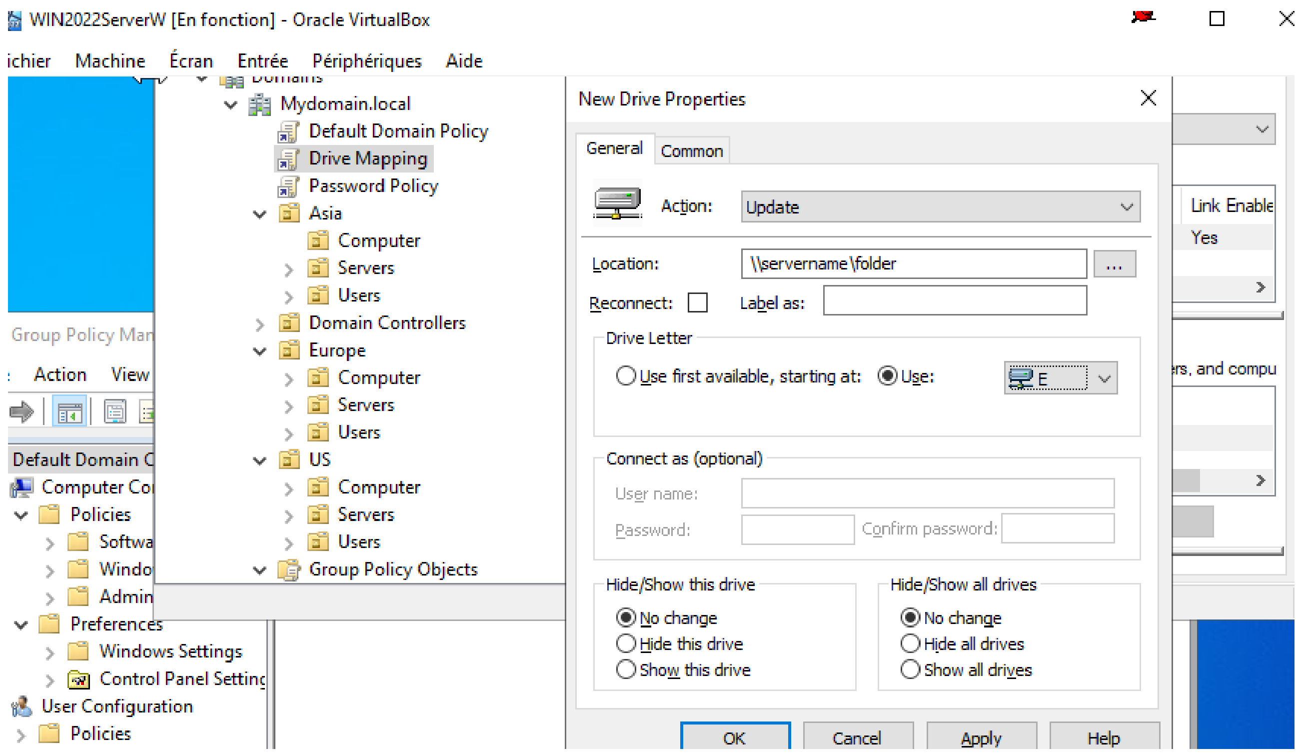

Figure 20: Configure the Drive Mapping Policy

Step 21 — Create a Control Panel Restriction Policy

What I did: I created another Group Policy Object to restrict access to Control Panel and related PC settings.

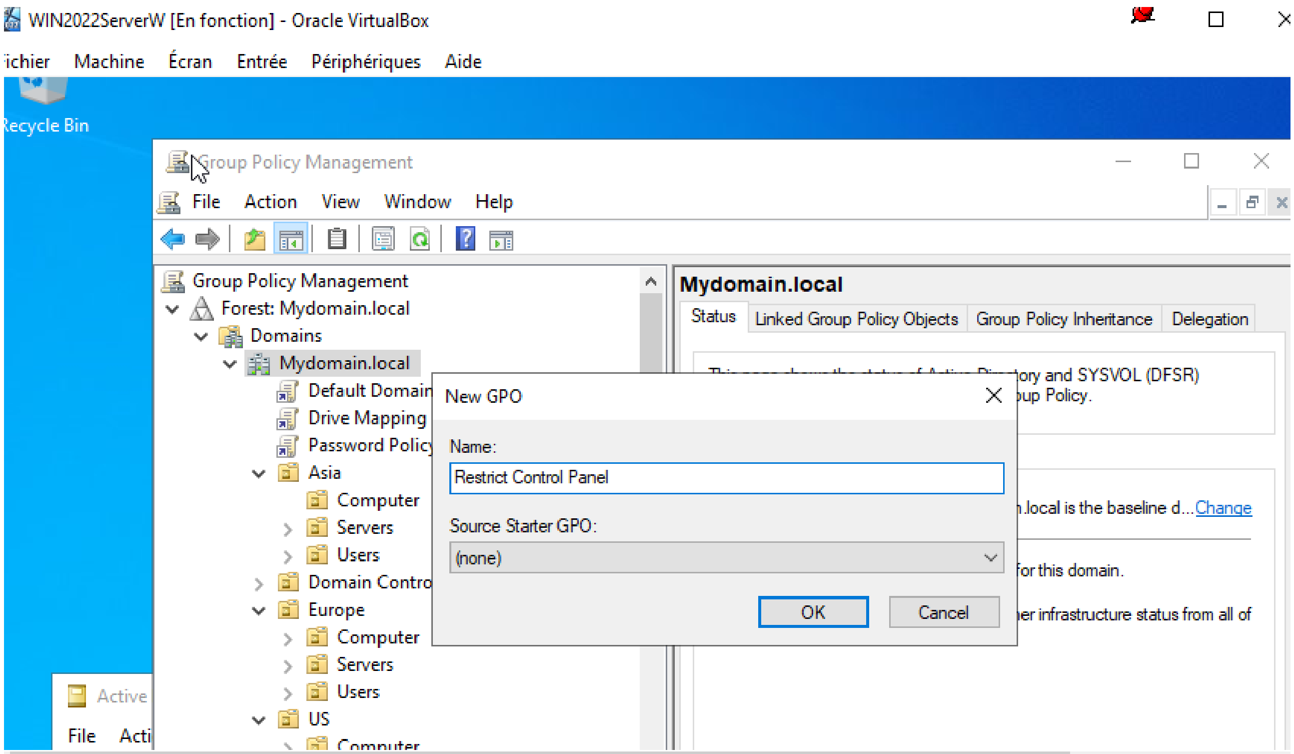

Figure 21: Create a Control Panel Restriction Policy

Step 22 — Enable 'Prohibit access to Control Panel and PC settings'

What I did: In the Group Policy Management Editor, I opened the setting Prohibit access to Control Panel and PC settings and enabled it.

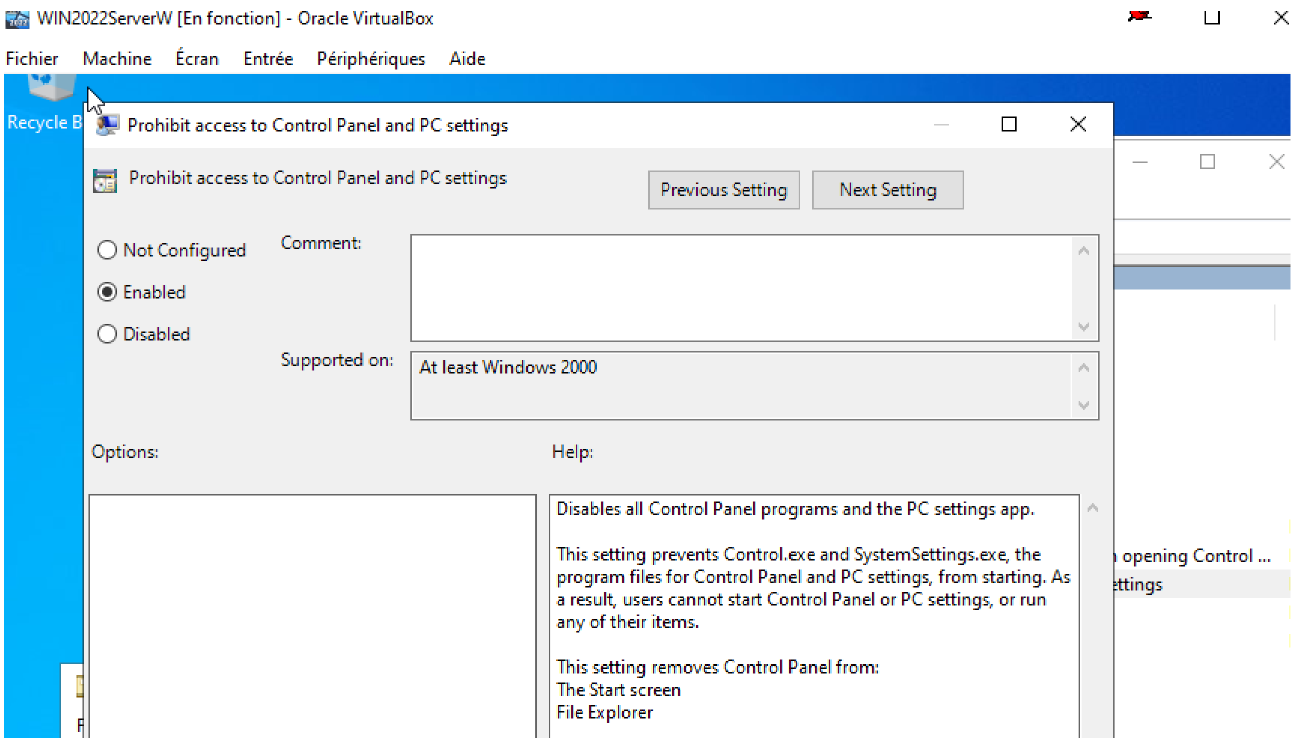

Figure 22: Enable 'Prohibit access to Control Panel and PC settings'

Step 23 — Create a Removable Storage Restriction Policy

What I did: I created/configured a Group Policy related to removable storage access and opened the corresponding policy setting.

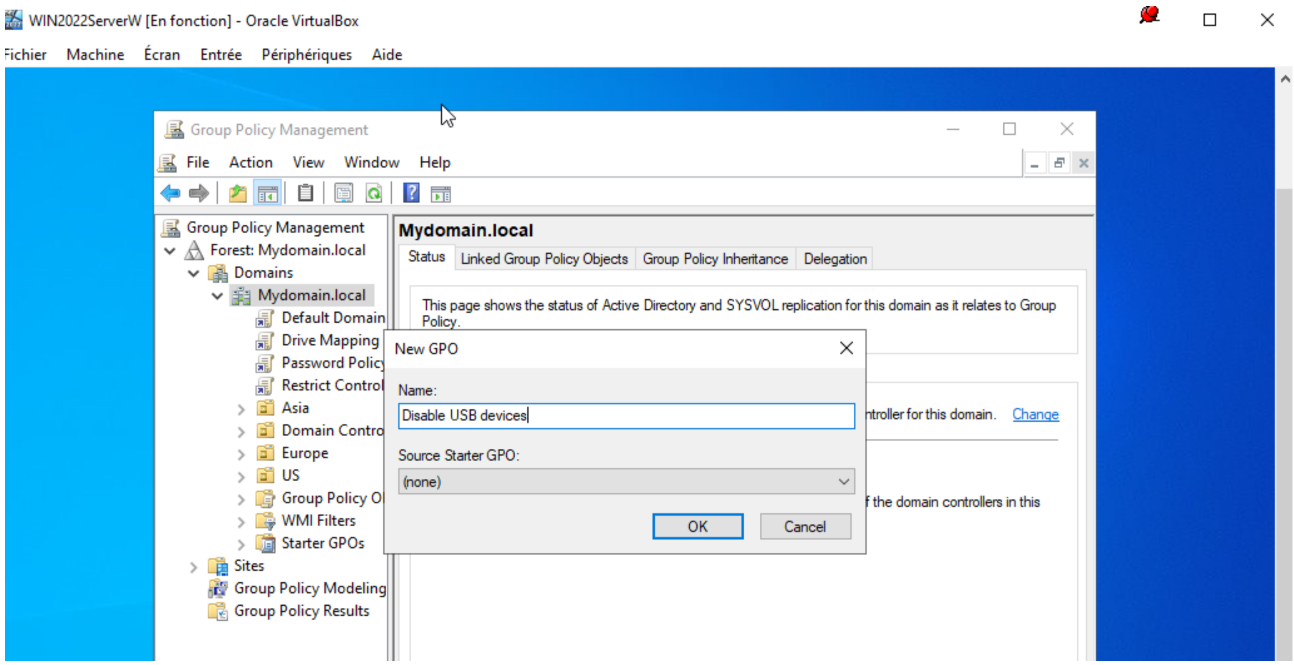

Figure 23: Create a Removable Storage Restriction Policy

Step 24 — Enable 'All Removable Storage Classes: Deny All Access'

What I did: I enabled the policy All Removable Storage classes: Deny all access to prevent access to removable-storage classes.

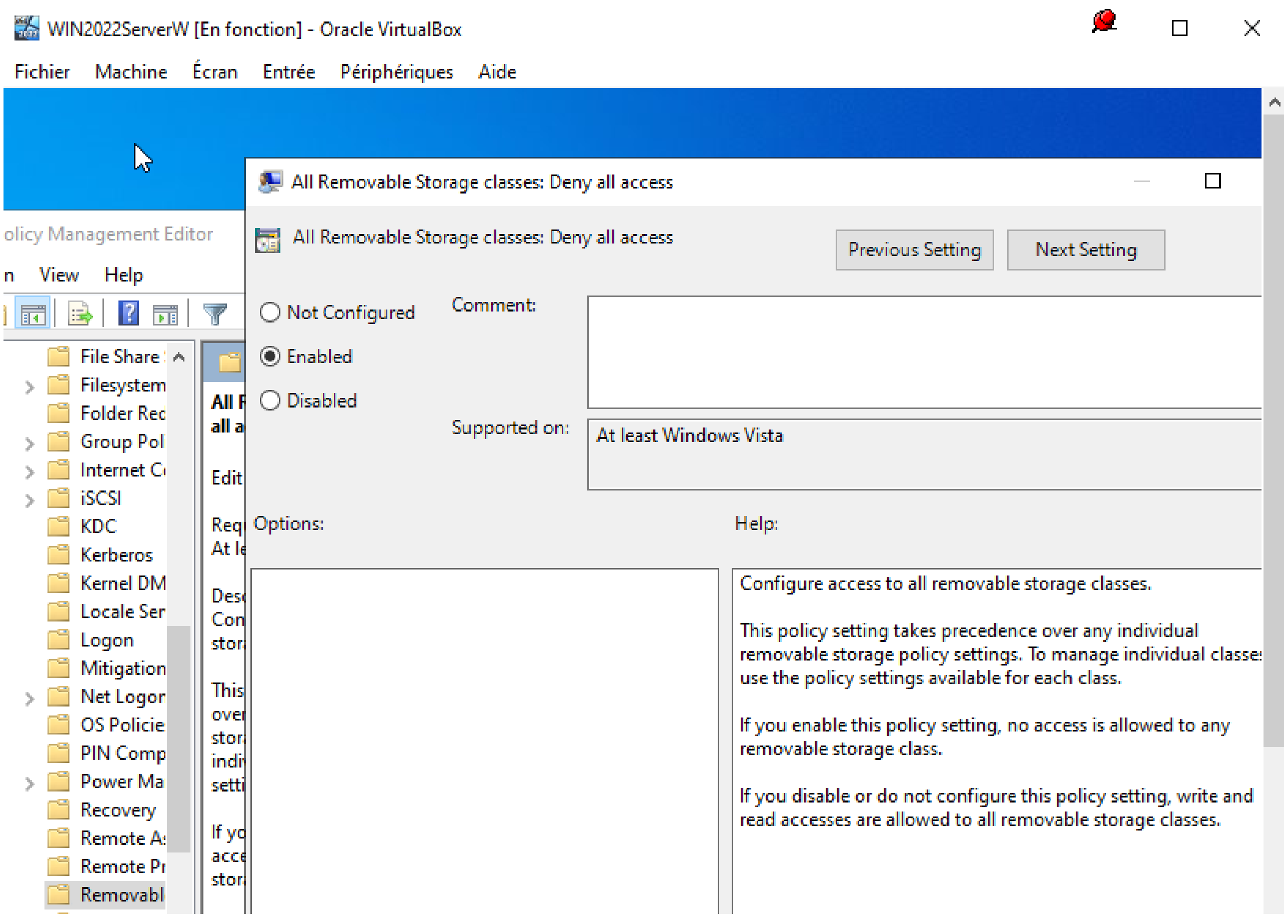

Figure 24: Enable 'All Removable Storage Classes: Deny All Access'

Step 25 — Prepare the Windows 10 Client

What I did: I prepared the Windows 10 client VM as the workstation that will join the Active Directory domain.

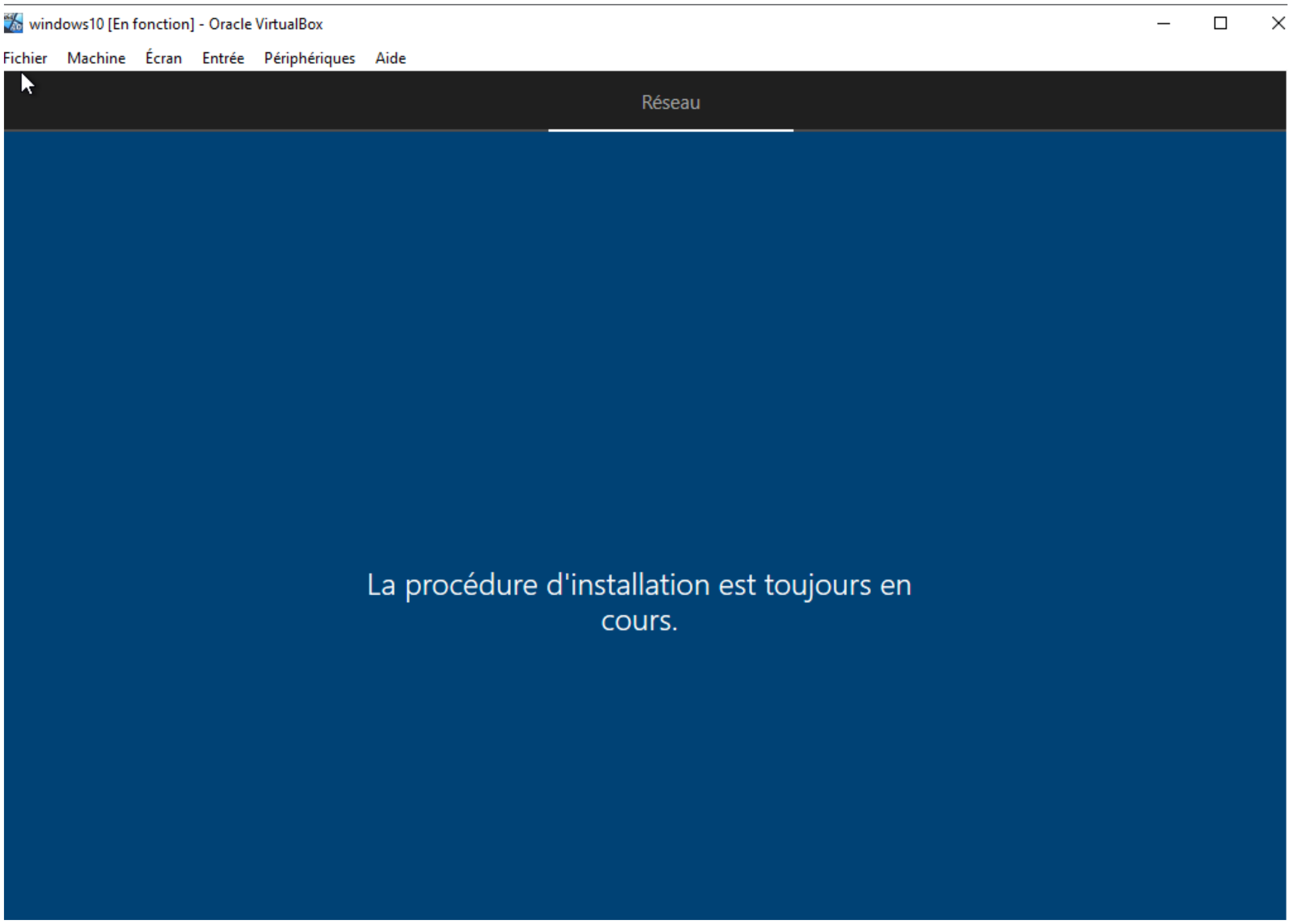

Figure 25: Prepare the Windows 10 Client

Step 26 — Configure the Windows 10 Client IPv4 and DNS Settings

What I did: I configured the Windows 10 client with a static IPv4 address. The screenshot shows 192.168.20.11/24 and the preferred DNS server 192.168.20.10.

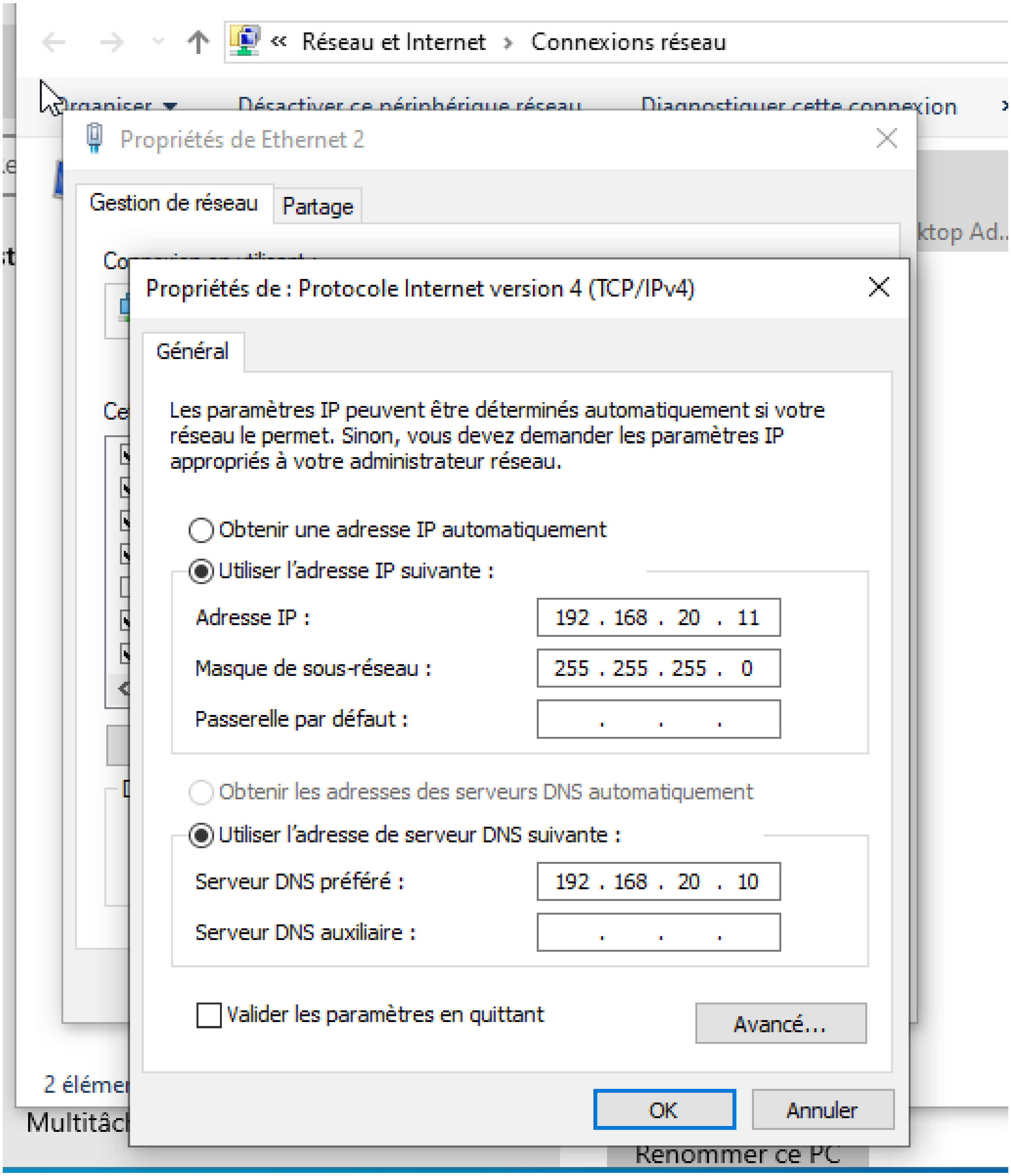

Figure 26: Configure the Windows 10 Client IPv4 and DNS Settings

Step 27 — Rename the Windows 10 Client to PC1

What I did: I opened the Windows 10 computer-name settings and renamed the client computer to PC1.

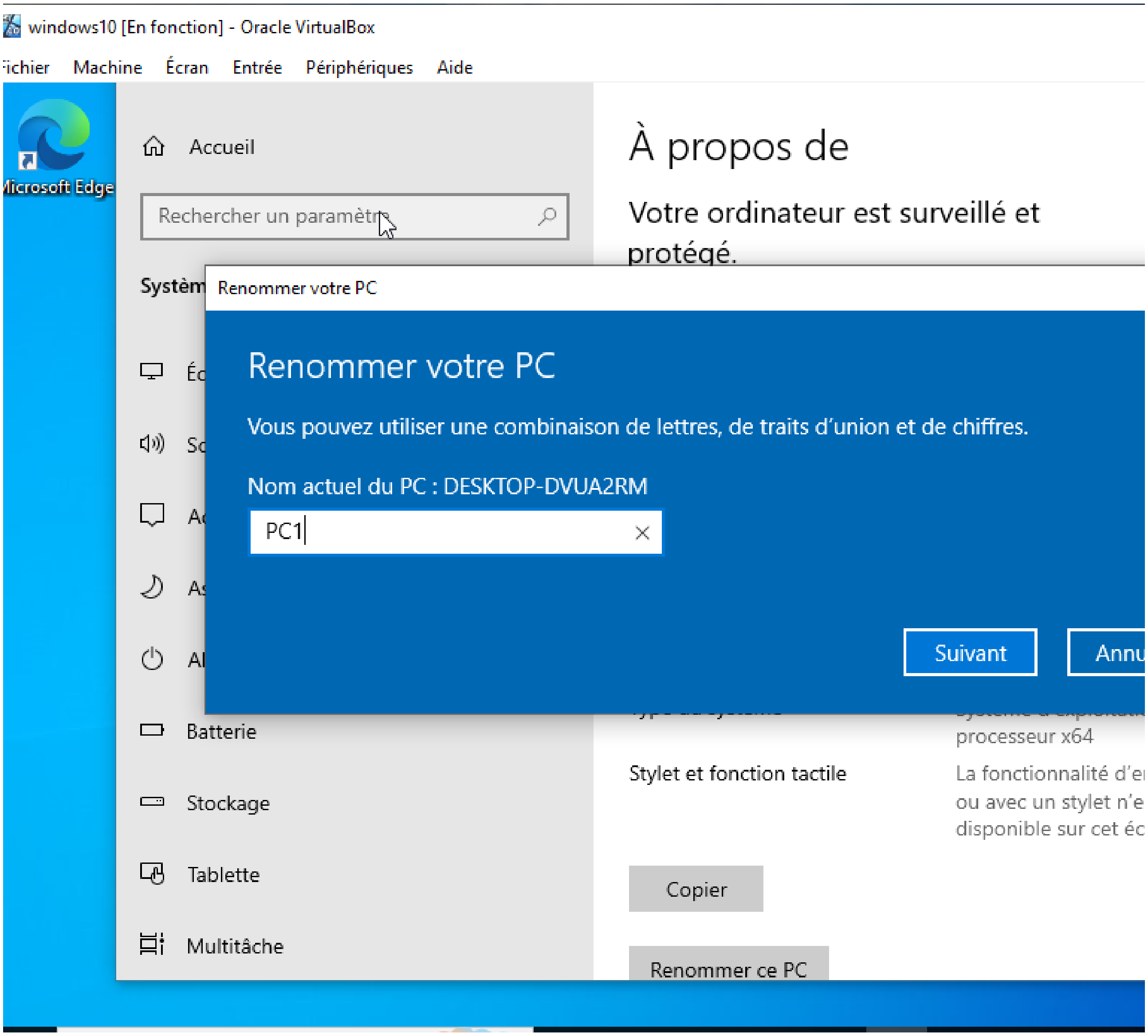

Figure 27: Rename the Windows 10 Client to PC1

Step 28 — Join Windows 10 to the Mydomain.local Domain

What I did: In System Properties, I changed the computer membership from WORKGROUP to the Mydomain.local domain.

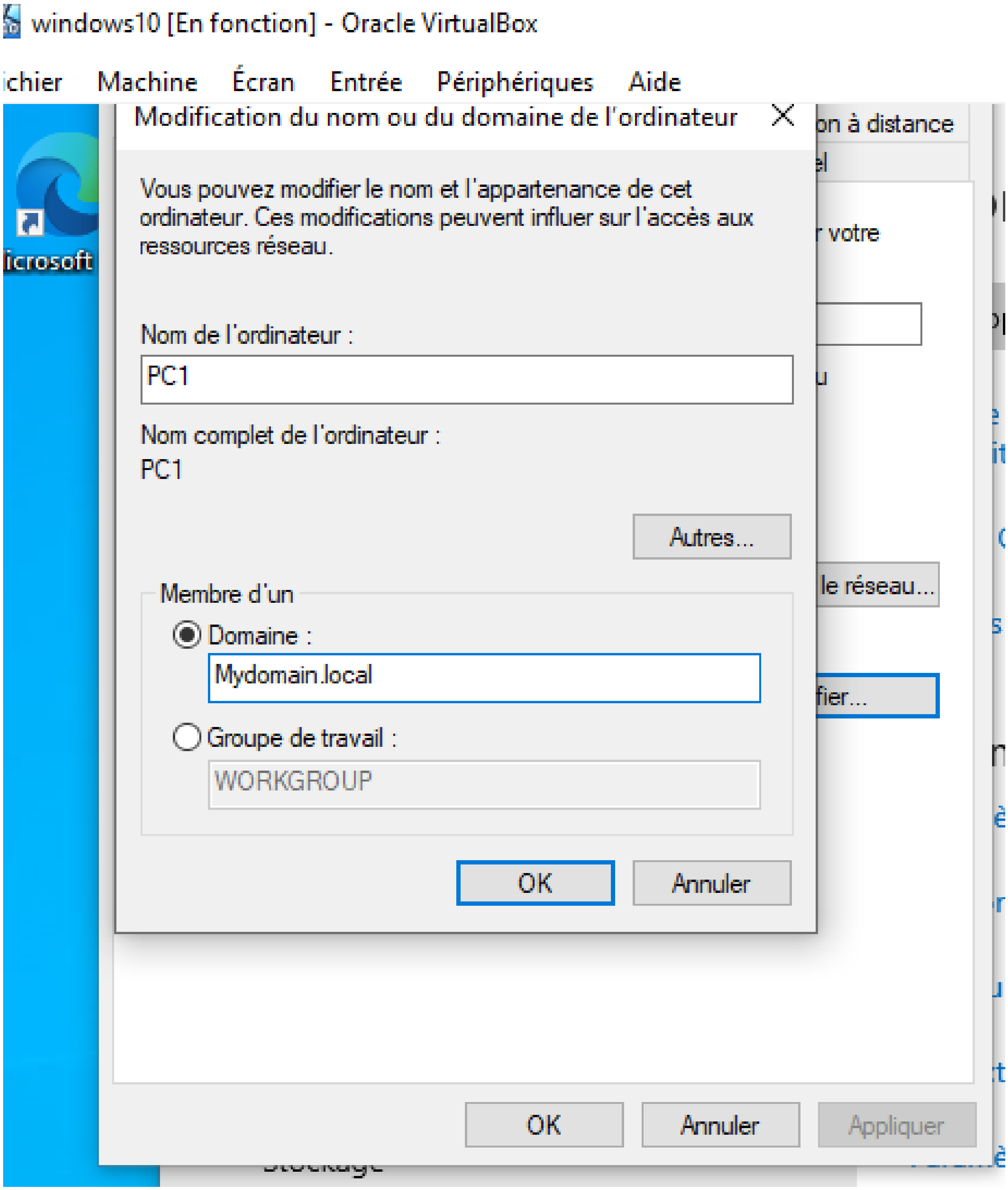

Figure 28: Join Windows 10 to the Mydomain.local Domain

Step 29 — Confirm Successful Domain Join

What I did: Windows confirmed that the computer successfully joined the Mydomain.local domain.

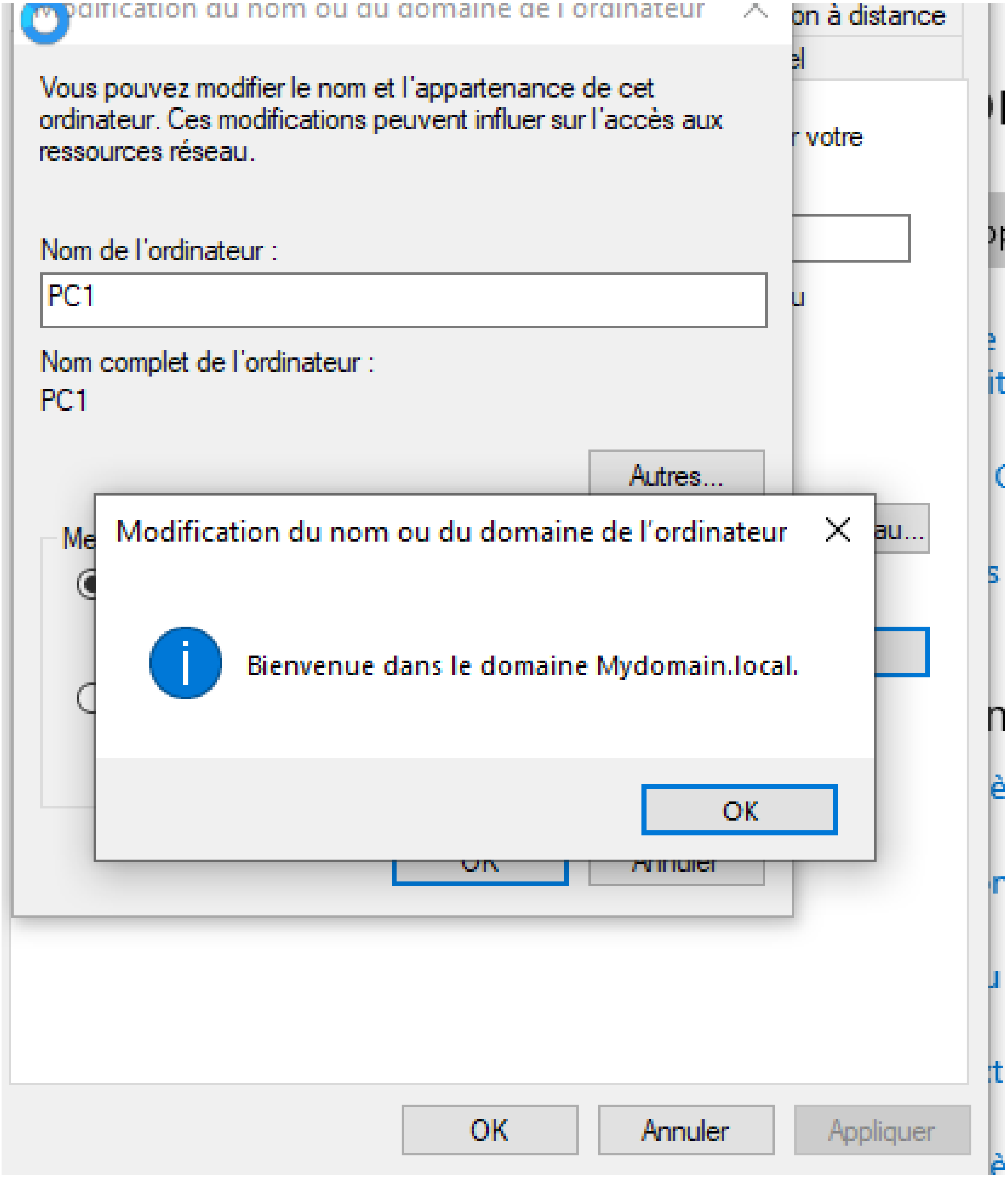

Figure 29: Confirm Successful Domain Join

Step 30 — Verify the Client Computer in Active Directory

What I did: Back on the Domain Controller, I opened Active Directory Users and Computers and verified that the client computer PC1 appears in Active Directory.

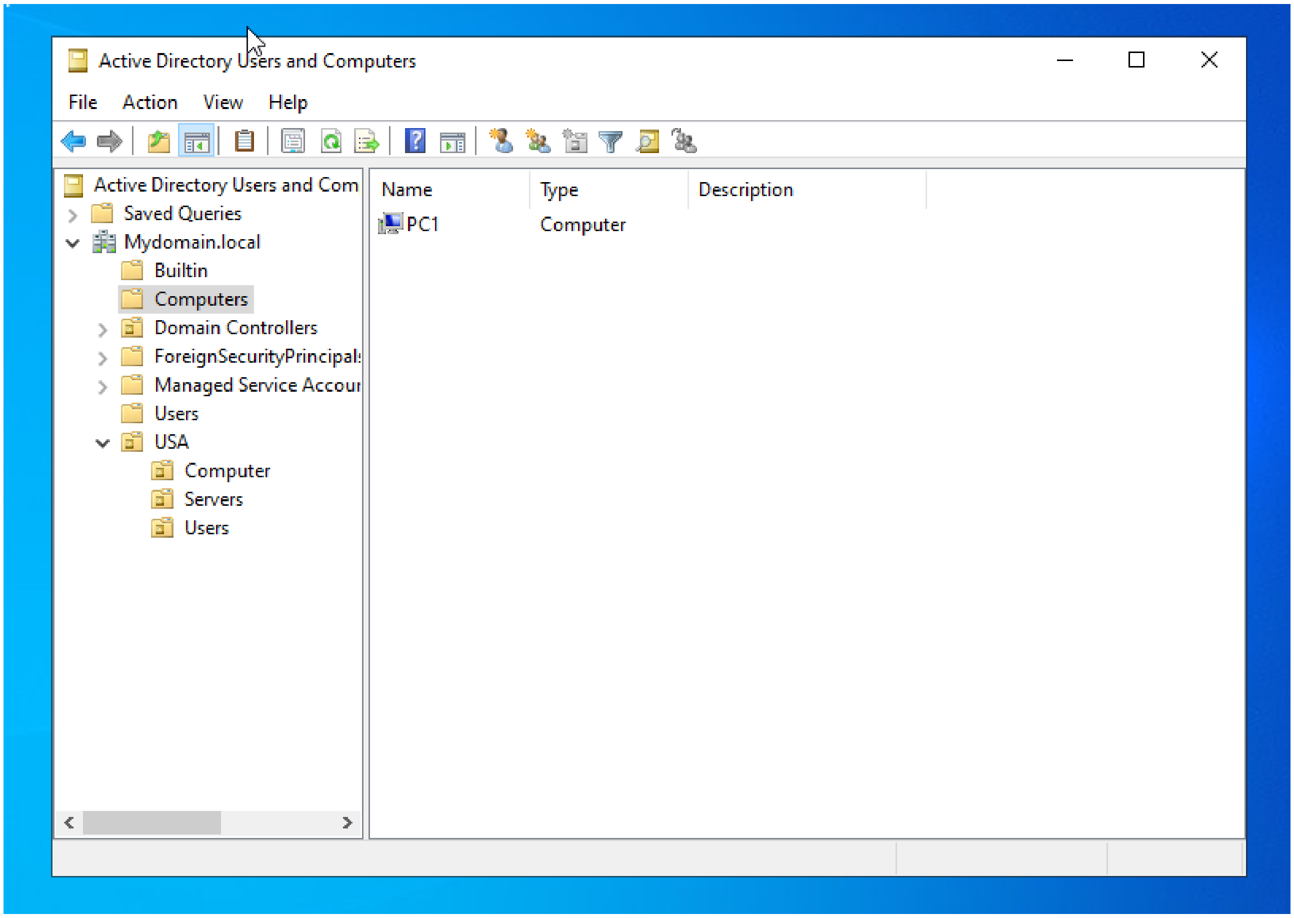

Figure 30: Verify the Client Computer in Active Directory

Step 31 — Log in as a Domain User

What I did: On the Windows 10 client, I selected Other user and began a domain-user sign-in using the MYDOMAIN domain.

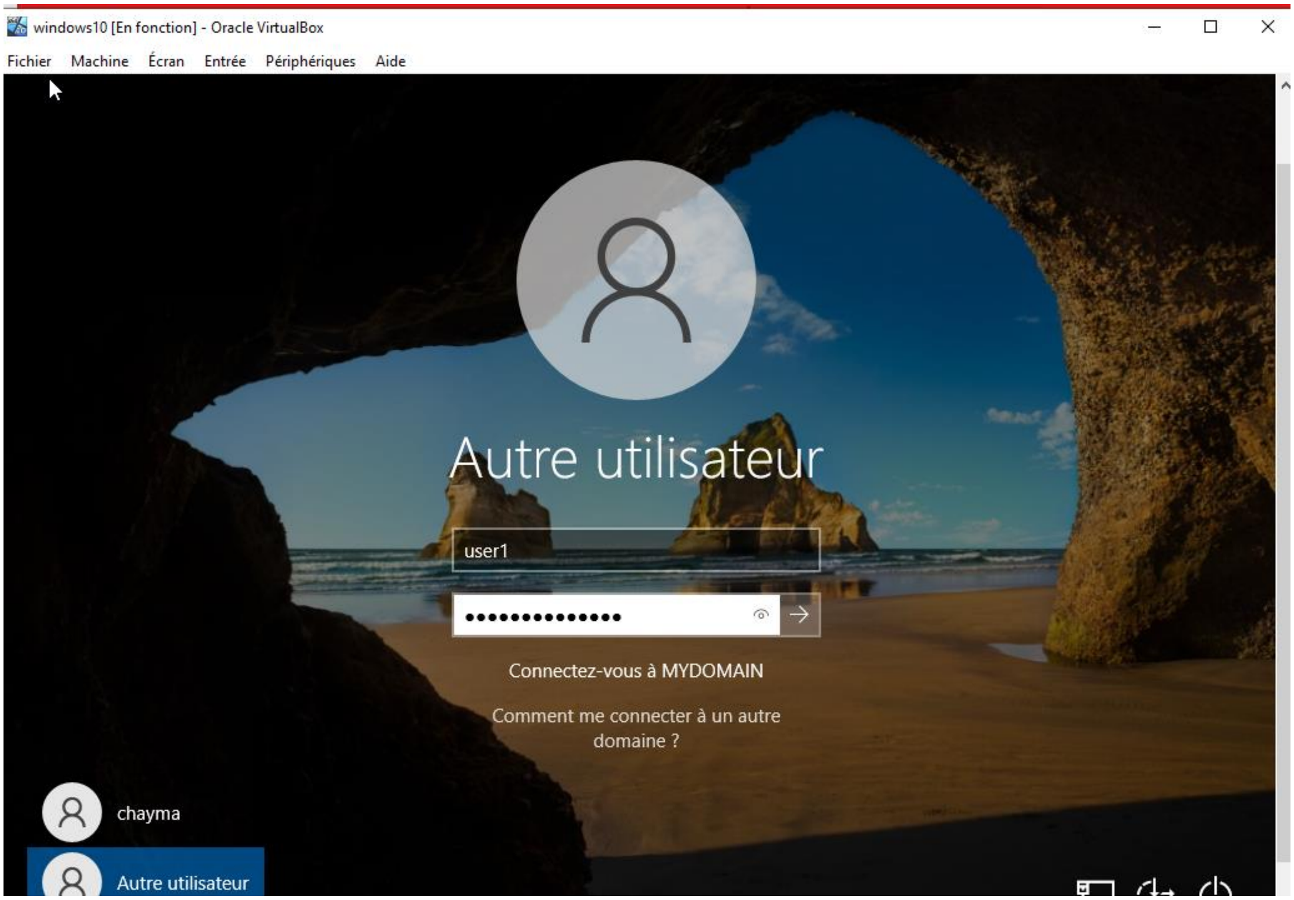

Figure 31: Log in as a Domain User

Step 32 — Sign in as User01

What I did: I successfully signed in to the Windows 10 client as the domain user User01.

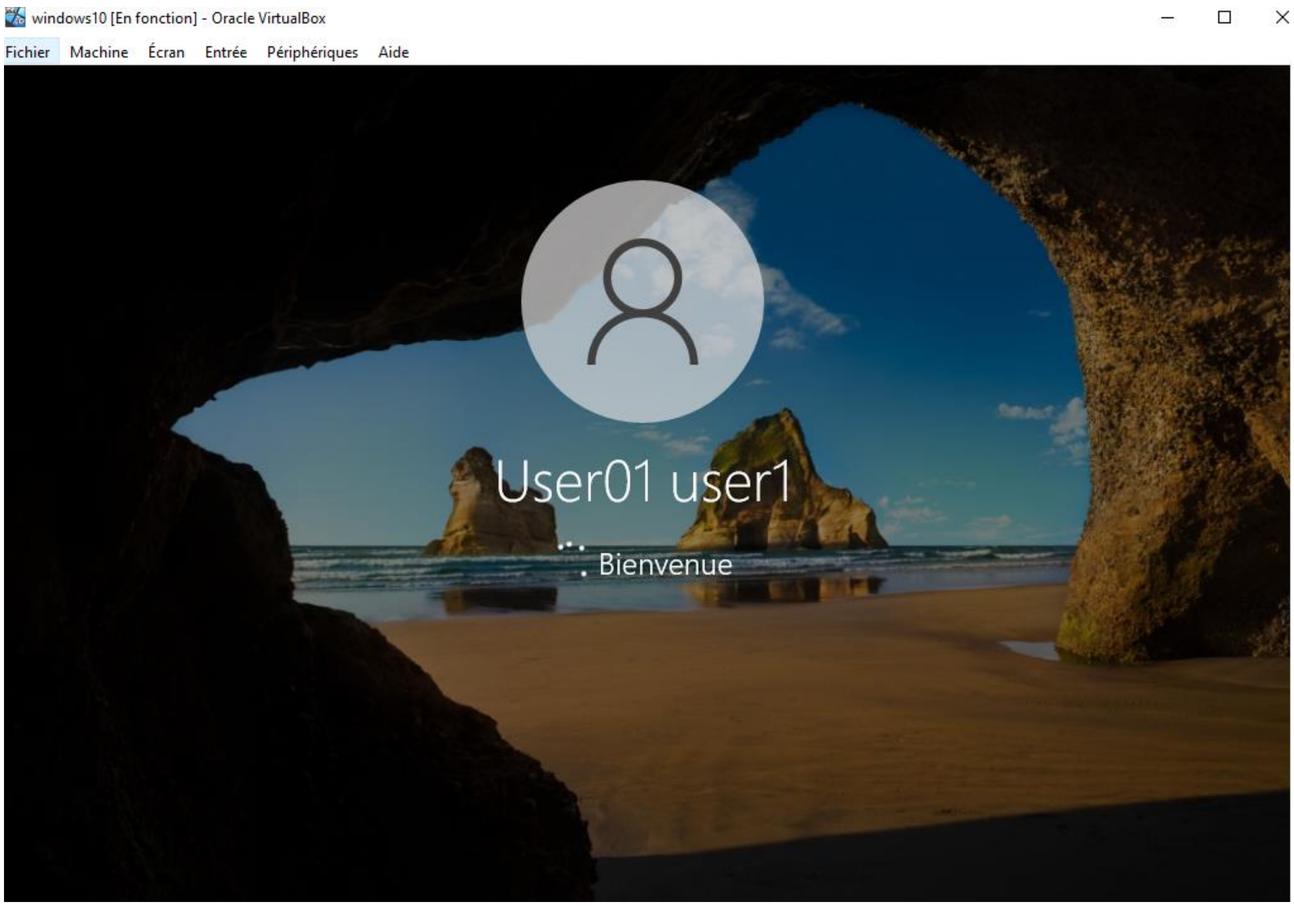

Figure 32: Sign in as User01